

The Compass



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**Community Mathematics Centre
Schools of the Krishnamurti Foundation India (KFI)**

Statement of intent

The magazine is published by the Community Mathematics Centre. It is a collaborative initiative of the schools of the Krishnamurti Foundation India (KFI).

Its intention is:

- To provide material for mathematics and science teachers who teach at the high school or middle school levels, and also for students who study at these levels.
- To serve as a platform where teachers and students can share their findings and discoveries with others who share their love for the subject.

In addition to mathematical content, in each issue we shall share a few ‘snippets’ from the world of science — short extracts abstracted from articles in online magazines. These will be selected on the basis of intrinsic interest or connection with the world of mathematics.

The Compass is published as an online magazine free of cost.

Articles may be downloaded and used freely by teachers, but with proper attribution.

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Colour coding

The magazine has articles addressed to teachers who teach at different levels: Lower Primary level; Upper Primary and High School levels; and Senior Secondary and Junior College levels. There is also a Problem Corner. In addition we have the Snippets.

For the convenience of the reader, we use the following colour coding to distinguish between the various levels. Of course, such a division can never be exact. So, our apologies in advance for errors of judgement!

- Primary & Upper Primary (Classes 4, 5, 6, 7)
- Upper Primary & High School (Classes 6, 7, 8, 9, 10)
- Senior Secondary & Junior College (Classes 10, 11, 12)
- Problem Corner
- Snippets

Introducing the magazine ...

In “Prelude to Mathematics” — a remarkable book written by a great writer and a great mathematics educator, W W Sawyer — the author declares his intentions in the very first chapter: “This is a book about how to grow mathematicians.” Of course, education is about “growing” human beings and not only about growing mathematicians! With that understanding in mind, it is worth delving deeper into what the author is telling us. We cannot do better than to quote from the first chapter of Sawyer’s book. There is much to learn from what he has written (even if one does not agree with everything he says).

This is a book about how to grow mathematicians. Probably you have no intention of trying to grow mathematicians. Even so, I hope you may find something of interest here. I myself have no intention of growing plants. ... But I like very much to look at gardens other people have grown. And I am still more interested if I can meet a man who will explain to me ... just how a plant grows, how, when it is a seed under the earth, it knows which way is up for its stem to grow and which way is down for the roots; how a flower manages to face towards the light; what chemical elements the plant needs from the soil, and just how it manages to rearrange them into its own living tissue. The interest of these things is quite independent of whether one actually intends to go out and do some hoeing.

What I am trying to do here is to ... understand what growth is. I am not writing for the professional teacher of mathematics ... but for the person who is interested in getting inside the mind of a mathematician.

It is very difficult to communicate the things that are really worth communicating. Suppose, for instance, that you have spent some years in a certain place, and that these years are particularly significant for you. They may have been years of early childhood or school days, or a period of adult life when new experiences, pleasant or unpleasant, made life unusually interesting. If you revisit this place, you see it in a special way. Your companions, seeing it for the first time, see the physical scene, a pleasant village, a drab town street, whatever it may be. They do not see the essential thing that makes you want to visit the place; to make them see it, you need to be something of a poet; you have to speak of things, but so as to convey what you feel about those things.

But such communication is not impossible. Generally speaking, we overestimate the differences between people. I am sure that if one could go and actually be somebody else for a day, the change would be much less than one anticipated. The feelings would be the same, but hitched onto different objects. Most human misunderstandings are due to the fact that people talk about objects and forget the varying significance the same object can have for different people.

Generally speaking, teaching conveys thoughts about objects rather than living processes of thought. Suppose someone comes to me with some kind of puzzle. It may be a question in a child's arithmetic book, or a serious problem of scientific research. Perhaps I succeed in solving the puzzle. Then it is quite easy to explain the solution. Suppose I do so; I have shown the question of how to deal with that particular problem. But if another problem of a different kind arises, I shall be consulted again. I have not made my pupil independent

of me. What would be really satisfactory would be if I could convey, not simply the knowledge of how to solve a particular puzzle, but the living attitude of mind that would enable my pupil to attack puzzles successfully without help from anyone.

In these few lines, Sawyer has captured his whole approach to teaching mathematics to children.

There are some remarkable lines in the above paragraphs. For example, when he writes that you need to be something of a poet to communicate the feeling of an experience to anyone. Or that we overestimate the differences between people, and that true communication is not impossible. These are extremely important observations and insights.

Elsewhere, in his widely-read book “Mathematician’s Delight” (and it is truly a delight to read), Sawyer writes, *Education consists in co-operating with what is already inside a child’s mind*. Later he writes, about the teaching-learning of geometry:

*The best way to learn geometry is to follow the road which the human race originally followed: **Do** things, **make** things, **notice** things, **arrange** things, and only then **reason** about things.*

We shall not discuss these views at this stage (there will be time enough for that, later), but we hope that everything in this magazine will carry the spirit of Sawyer’s words.

Fast Calculations – 1

In this series of articles, we shall study various computational shortcuts. Our approach will be to offer a series of examples with no explanation, and then ask you to figure out “what is going on.” So here we go!

Computing squares

Let’s see if you can figure out the algorithm from the following examples. (You may at some time have hit upon the procedure yourself.) In each case we show the working without any explanation. That’s up to you to find!

To find the square of 12

- $12 + 2 = 14$
- $2^2 = 4$
- Hence $12^2 = 144$

To find the square of 13

- $13 + 3 = 16$
- $3^2 = 9$
- Hence $13^2 = 169$

To find the square of 14

- $14 + 4 = 18$
- $4^2 = 16$
- Hence $14^2 = 196$ (note the role played by 1 in the tens place of 16)

To find the square of 15

- $15 + 5 = 20$
- $5^2 = 25$
- Hence $15^2 = 225$ (note the role played by 2 in the tens place of 25)

To find the square of 19

- $19 + 9 = 28$
- $9^2 = 81$
- Hence $19^2 = 361$ (note the role played by 8 in the tens place of 81)

What is the explanation for this shortcut?

For which numbers will it work?

For which numbers will it not work? What can be done to “fix” the procedure in such a situation?

Pretty Number Patterns – 1

In this series of articles, we showcase various pretty patterns from the world of numbers. There is no dearth of such patterns!

As in the previous article, our approach will be to offer a series of examples with no explanation, and then to ask you to figure out “what is going on.” So, here we go ...

Mischievous fractions

In this instalment we shall look at “mischievous fractions.” You may be familiar with the following equality:

$$\frac{16}{64} = \frac{1}{4}.$$

In the fraction on the LHS, we have ‘illegally’ cancelled the 6 which occurs both in the numerator and in the denominator. Of course, we are not permitted to do such a cancellation. We would incur a big penalty in an examination or a homework assignment if we made such an error!

But to our surprise, no arithmetic error has occurred here; $16/64$ is indeed equal to $1/4$; please check! So we can write:

$$\frac{1\cancel{6}}{\cancel{6}4} = \frac{1}{4}.$$

Even more surprising: infinitely many instances of mischievous cancellations can be generated from this example! Thus we have:

$$\frac{166}{664} = \frac{1}{4}, \quad \text{i.e.,} \quad \frac{1\cancel{6}6}{\cancel{6}64} = \frac{1}{4},$$

and:

$$\frac{1666}{6664} = \frac{1}{4}, \quad \text{i.e.,} \quad \frac{1\cancel{6}66}{\cancel{6}664} = \frac{1}{4},$$

and:

$$\frac{16666}{66664} = \frac{1}{4}, \quad \text{i.e.,} \quad \frac{1\cancel{6}666}{\cancel{6}6664} = \frac{1}{4},$$

and so on, for ever and ever!

That is, we have:

$$\frac{1}{4} = \frac{16}{64} = \frac{166}{664} = \frac{1666}{6664} = \frac{16666}{66664} = \dots$$

How would you explain these equalities?

What is happening “underneath the surface” that makes these cancellations possible?

See if you can find more instances of such mischievous fractions. To do this you would have to answer the question posed above: what exactly is happening that makes these cancellations possible?

We leave the full exploration to you!

Closing remark

In this article and the earlier one, we gave examples of computational shortcuts and number patterns. These form attractive topics for exploration and study. Their common feature is that both of them offer excellent entry points into the study of algebra.

We shall continue these explorations in future issues of the magazine.

Discovering the Triangle Inequality

Article contributed by Shivkumar K M & Shravan S K

The school bell rings! Everyone is sprinting out of the school gates with heavy bags on their backs. Anvesh meets Sumedha, his cousin, on his way home and expresses his disgust about maths and also about the high standards set at the Kendriya Vidyalayas (KV), his school. He struggles at math. Sumedha is passionate about math and she wants to kindle Anvesh's interest towards the subject.

The Exploration

Sumedha I will give you 3 numbers. Can you draw a triangle whose sides have lengths as the given numbers?

Anvesh Okay, that seems easy. I will do it.

Sumedha The numbers are 3, 4, 6.

(Anvesh takes out his pencil, scale, and compass to draw the triangle. He draws a straight line of length 4 units. Using the compass, he draws 2 arcs having radius 3 centred at each corner of the straight line. Again, he draws arcs of radius 6 similarly. He joins an intersection of the arcs with the ends of the straight line.)

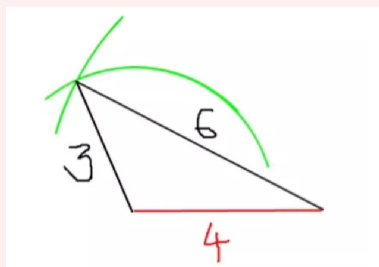


Figure 1

Anvesh Whoo! See, I did it.

Sumedha Great! Try 7, 7, 7.

Anvesh Done.

Sumedha Ok, now try 5, 4, 11.

(Anvesh tries many times and he is not able to construct a triangle out of it.)

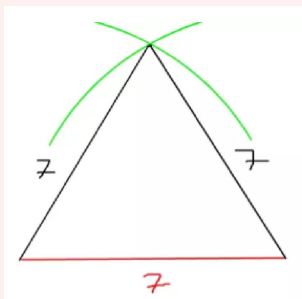


Figure 2

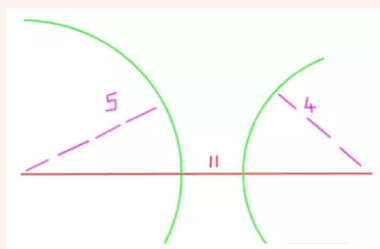


Figure 3

Anvesh I am not able to do it. I think something is wrong with these numbers.

Sumedha Okay. Let us see what has gone wrong. Why are you not able to draw the triangle using 5, 4, 11?

Anvesh Umm ..., Because the arcs are not meeting (intersecting) at any point although I drew complete circles. I don't have the third vertex of the triangle.

Sumedha Yes, right. So, why are the arcs not meeting anywhere?

Anvesh (Looks at his paper, after a while) I don't know.

Sumedha I will give you another set of numbers: 4, 6, 10. Draw a triangle for these numbers.

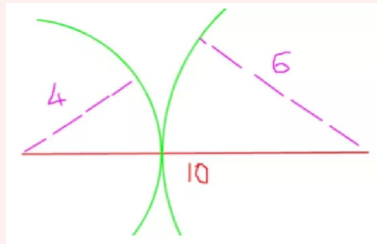


Figure 4

Anvesh Uh-oh! Again, something is wrong with these numbers too. Couldn't get the triangle.

Sumedha (Looking at his paper) So, why couldn't you draw the triangle? There is a point where the arcs have met.

Anvesh Because, the intersection point lies on the straight line. If all three points are on a straight line, then the triangle can't be constructed.

Sumedha Good! You're smart.

Anvesh Thanks, Summi!

Sumedha Now you have learnt that we cannot always draw a triangle given any 3 numbers. If I give you 3 numbers, can you say if it is possible to draw a triangle without drawing it on the paper?

Anvesh Oh, that is not possible. I can say if it is possible by drawing it on the paper.

Sumedha No, drawing is not allowed.

Anvesh (After thinking for a while) Can you give me some clue?

Sumedha Hmm, okay. Have a look at the triangles you were able to draw and the ones you couldn't. See if you can find some common facts in the triangles you drew.

Anvesh The arcs always meet & such points should not lie on the line.

Sumedha Nice! Now, tell me about the triangles you weren't able to draw.

Anvesh Either the arcs don't meet anywhere or the arc meeting point lies along/on the straight line.

Sumedha Good, so look into the diagram of 4, 6, 10 (Figure 4).

Anvesh The arcs intersect on the straight line.

Sumedha Yes. Why do you think that is happening? Why do they intersect on/along the line?

Anvesh (Does some thinking and arithmetic) As the largest of the 3 numbers is equal to the sum of the other 2 numbers, the arcs (or circles) meet along/on the straight line.

Sumedha Very good, you are halfway to finding the answer. Now, in the 5, 4, 11 drawing, why didn't the arcs meet anywhere even though you drew full circles? What is the problem with them?

Anvesh I think if the radius of either of the circles were a bit longer, they would intersect at a point. Am I right?

Sumedha Absolutely. That's correct. If the side length 4 is increased to 7, can you draw a triangle?

Anvesh I guess so. Let me see. (After a while) Done.

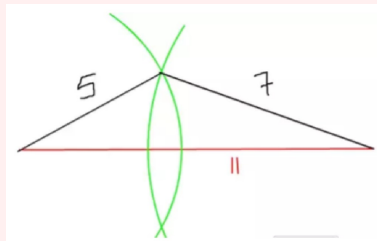


Figure 5

Sumedha Can we also say that as the sum of the two smaller numbers is less than the largest one (5, 4, 11 in Figure 3) we couldn't construct the triangle?

Anvesh Hmm ... (Peeks at his drawing). I think yes.

Sumedha Okay, so now will you tell me if it is possible to draw a triangle if I give you 3 numbers?

Anvesh I will try.

Sumedha You can do it. For 2, 3, 4?

Anvesh Yes, it can be drawn. Am I right?

Sumedha You are right! Can you explain?

Anvesh $2 + 3 (= 5)$ is greater than 4. So, the arcs meet at some point outside the line. $2 + 3 (= 5)$ is not equal to 4. So, the arcs' intersection point doesn't lie on/along the straight line. Hence, we can connect the ends of the line and the intersection point to form a triangle.

Sumedha Correct. Try 8, 5, 13.

Anvesh Not possible. $8 + 5 = 13$. The arcs intersect on/along the line.

Sumedha Right. What about 9, 23, 12?

Anvesh Not possible. $9 + 12 (= 21)$ is less than 23. The arcs do not intersect anywhere.

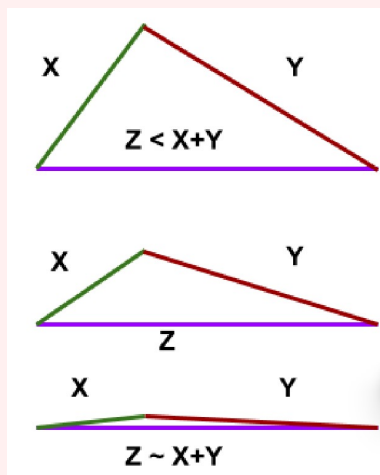


Figure 6

Sumedha (See Figure 6.) Keeping Z constant, see how the triangle changes on decreasing X and Y (i.e. the sum $X + Y$ gets lesser and closer to Z). When $X + Y = Z$, then the point of intersection lies on the line.

Sumedha For any given 3 numbers, if one of the 3 conditions fails (see Figure 7), then the triangle can't be constructed using those numbers. This rule is well-known as Triangle Inequality. From the constructions we have seen that if the Triangle Inequality holds, then a triangle is possible; if the Triangle Inequality doesn't hold, then a triangle is not possible."

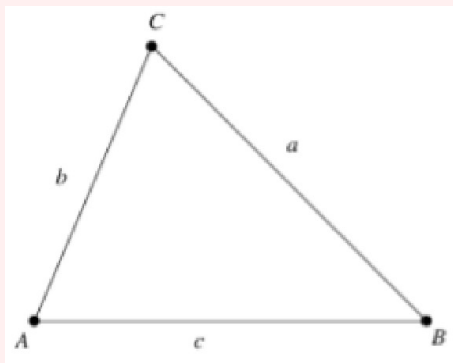


Figure 7: $a + b > c$, $b + c > a$, $c + a > b$

Anvesh Thanks a lot Summi!

Sumedha You're welcome Annu.

Anvesh Ok then, I will leave now!

Sumedha Okay. One thing before you leave. Now that you have cracked a problem, do you want to try your hand at solving another?

Anvesh Of course! This seems so exciting!! Can you tell me the problem?

Sumedha We have seen how given three numbers, one can determine the existence of a triangle having the numbers as sides. If I give you four numbers and ask you if there exists a quadrilateral with these numbers as sides, will you be able to answer this without drawing?

Anvesh Hmm, I am curious to find out. I will try it out. Good bye!

Anvesh walks home joyfully satisfied with his discovery & wondering about the quadrilateral problem.

Like Anvesh, we hope that our readers also will give the quadrilateral problem a thought!

Remark

The name Anvesh is derived from the Sanskrit word *Anveshak*, meaning “Seeker” (here, knowledge seeker), and the name Sumedha is derived from the Sanskrit word *Sumedhas* meaning “Intelligent.”

A discussion on the children’s analysis

While it may be intuitive that the sides of a triangle satisfy the triangle inequality (a direct straight route from A to B is shorter than the route of going from A to a point C lying in a different direction, and then going from C to B), the fact that if three numbers satisfy the inequality, then a triangle exists with these numbers as sides is not something that is apparent. The above discussion between Sumedha and Anvesh shows that the triangle inequality is also a sufficient condition for the existence of a triangle.

Also, while considering triangle constructions, Anvesh and Sumedha were mostly taking the longest side as the base. Can’t a triangle be constructed if either of the other sides are taken as base? Of course, they can be constructed. For example, constructing the triangle with sides 5, 7 and 11 with the base having length 5 will look as it appears in Figure 8.

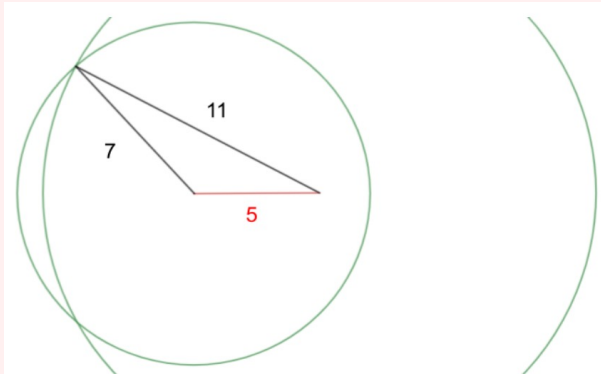


Figure 8

If we consider triangle construction with one of the shorter sides as the base, will we still get the triangle inequality as the necessary and sufficient condition for the existence of the triangle?

The great physicist J. Gibbs once remarked “One of the principal objects of theoretical research is to find the point of view from which the subject appears in the greatest simplicity.” Among the two ways of viewing the construction of triangles as discussed above, can you see under which view-point the problem appears in the greatest simplicity?

Making the students discoverers

Instead of stating the Triangle Inequality Rule at first and giving examples, the above approach gives students an opportunity to question themselves (Why so? How to do that? Why not? etc.), think and hopefully discover the answer. By undertaking such an exploration many students ACTUALLY

understand what is going on and they cherish the process.

About the authors ...

This exploration was facilitated by Shiv & Shravan with some students of a Karnataka State Govt. School as part of weekly math mentorship, an initiative of NoteBook Drive, Indian Institute of Science.

Shiv has completed his PhD from the ECE, IISc. Shravan has completed his MSc (Engg.) from Dept. of CDS, IISc. Both are now part of Seed2Sapilng Education working towards making math joyful & enriching for both students & teachers, and are happy to collaborate with like-minded people to improve the quality of math education in the country.

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Puzzles to Puzzle You: Crazy Cryptarithms

The answers to these cryptarithms are given on page 105. Solutions will be given in the next issue of the magazine.

Crazy Cryptarithm – 1

Solve the following cryptarithm:

$$\begin{array}{r} A \ B \ C \ D \\ \times \qquad \qquad 4 \\ \hline D \ C \ B \ A \end{array}$$

Crazy Cryptarithm – 2

Solve the following cryptarithm:

$$\begin{array}{r}
 A\ B \\
 +\ B\ C \\
 +\ C\ A \\
 \hline
 A\ B\ C
 \end{array}$$

Crazy Cryptarithm – 3

Solve the following cryptarithm, given that $A = 1$:

$$\begin{array}{r}
 B\ A\ D \\
 \times\ \quad\ 9 \\
 \hline
 G\ O\ O\ D
 \end{array}$$

Crazy Cryptarithm – 4

Solve the following cryptarithm (it is quite challenging!):

$$\begin{array}{r}
 R\ O\ U\ N\ D \\
 \times\ \quad\ \quad\ 3 \\
 \hline
 S\ Q\ U\ A\ R\ E
 \end{array}$$

Adding by Folding

Article contributed by Padmapriya Shirali

Gauss's method of adding the numbers from 1 to 100 is well-known. You may have come across it in primary or upper primary school. Here's how the story goes. Gauss's teacher set a task to the students (hoping to keep them busy!): add up the numbers from 1 to 100. Gauss quickly gave the answer:

$$1 + 2 + 3 + \cdots + 98 + 99 + 100 = 5050.$$

How did he get the answer so quickly? He saw that if the series is folded in the middle, the pairs have equal sums: $1 + 100 = 101$, $2 + 99 = 101$, $3 + 98 = 101$, and so on. Each pair sums to 101. Please check!

As there are 50 such pairs, the total must be $50 \times 101 = 5050$.

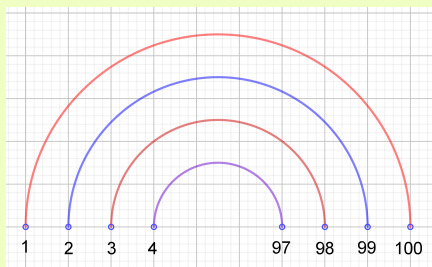


Figure 9: Adding $1 + 2 + \cdots + 99 + 100$

Variations of the problem Students can tackle variations of this problem by using the same approach. Here are some possible questions:

1. What is the sum of the numbers from 1 to 10? From 1 to 20? From 1 to 30?
2. What is the sum of the numbers from 100 to 150? From 100 to 200? From 200 to 1000?

Seeking a generalisation

Students can arrive at a general formula by generalising the process of summation of pairs.

- In the first case (1 to 10), the number of pairs is 5. Each pair adds up to 11:

$$1 + 10, \quad 2 + 9, \quad 3 + 8, \quad 4 + 7, \quad 5 + 6.$$

- In the second case (1 to 20), the number of pairs is 10. Each pair adds up to 21.
- In the third case (1 to 30), the number of pairs is 15. Each pair adds up to 31.

We present the results in tabular form as follows.

Numbers	Number of pairs	Sums of pairs	Required total
1 to 10	5	11	55
1 to 20	10	21	210
1 to 30	15	31	465
1 to 40	20	41	820

Let us denote the number of integers by n .

To start with, suppose that n is an even number.

Then the number of pairs is $1/2$ of n . The sum of each pair is 1 more than n .

The required total is (number of pairs $\times$ the sum of each pair). Therefore the required total is

$$\frac{1}{2} \times n \times (n+1) = \frac{n(n+1)}{2}.$$

What happens when n is an odd number?

What is the sum of numbers from 1 to 15? One way of getting around the fact that there are now an odd number of numbers is to include 0 in the list. Naturally, the total does not change when we include 0.

The question we now ask is: what is the sum of numbers from 0 to 15? Notice that we have an even number of numbers in the list.

How many pairs can be formed with equal sum? 8 (We treat 0 as a partner for 15.)

What is the sum of each pair? 15

Hence the required total is $8 \times 15 = 120$.

What happens in the general case? We want the sum of the numbers from 0 till n ; here n is odd, so the number of numbers is $n+1$. The number of pairs that can be formed with equal sum is $1/2 \times (n+1)$. The sum of each pair is n . Hence the required total is

$$\frac{1}{2} \times (n+1) \times n = \frac{n(n+1)}{2}.$$

We get the same formula in both cases!

How would we apply this principle to compute the sum for integers from 11 to 20?

How many pairs of integers in this range? 5. Note that $5 = (20 - 10)/2$.

What is the sum of each pair? 31

What is the total sum? $31 \times 5 = 155$

Can the students create a general formula for this situation?

It can be $1/2 \times (\text{upper limit} - \text{lower limit}) \times \text{sum of the pairs}$.

The answer can also be obtained by subtracting the sum of integers from 1 to 10 from 1 to 20:

$$\frac{20(20+1)}{2} - \frac{10(10+1)}{2} = 210 - 55 = 155.$$

Is there any other approach? Try out different ways!

Summing the even numbers

How would they apply this principle for the sum of the even numbers from 0 to 20?

How many even integers are there? 2, 4, 6, 8, 10, 12, 14, 16, 18, 20. The number of even integers is $20/2 = 10$.

What are the pairs? 2 + 20, 4 + 18, 6 + 16, 8 + 14, 10 + 12

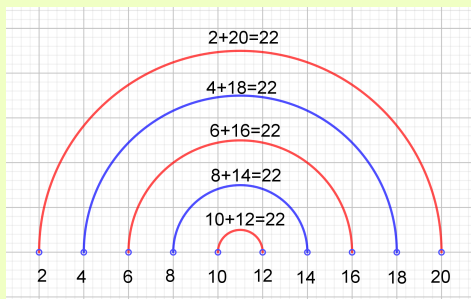


Figure 10: $2 + 4 + \dots + 18 + 20 = ?$

How many pairs? 5

What is the sum of each pair? $22 = 2 \times 11$

What will be the total sum? $5 \times 2 \times 11 = 10 \times 11 = 10 \times (10 + 1)$

What will be the general formula for this situation? If the number of even positive integers is n , the sum will be $n(n + 1)$.

Will the formula be the same for the sum of the odd numbers from 0 to 20?

How many odd integers are there in this series? $20/2 = 10$

What are the pairs here? $1 + 19, 3 + 17, 5 + 15, 7 + 13, 9 + 11$

How many pairs? 5

What is the sum of each pair? $20 = 2 \times 10$

What will be the total sum? $5 \times 2 \times 10 = 10 \times 10 = 10^2$

What will be the general formula for this situation? If the number of odd positive integers is n , the sum will be $n \times n = n^2$.

The formula is not the same!

Can the students figure out the sum of even integers from 21 to 50?

An exploration

Let us now look at the sums of sets of integers ten at a time.

Numbers	Sum	Difference in consecutive sums
1 to 10	55	
1 to 20	210	155
1 to 30	465	255
1 to 40	820	355

Is there a pattern in the way the sums are increasing?

By what quantity are the numbers increasing in the third column?

How do we explain this pattern?

Sums of multiples of 2

Let us now look at the sums of even integers in sets of integers ten at a time. How are the numbers increasing in the third column? How do we explain it?

Even numbers	Sum	Difference in consecutive sums
1 to 10	30	
1 to 20	110	80
1 to 30	240	130
1 to 40	420	180
1 to 50	650	230

Will such a constant increase happen if we consider the odd integers?

Odd numbers	Sum	Difference in consecutive sums
1 to 10	25	
1 to 20	100	75
1 to 30	225	125
1 to 40	400	175
1 to 50	625	225

Can the students prove that the increase in the total will always be 50 in the above case?

Sums of multiples of 3

Let us now look at the sums of multiples of 3 in sets of integers ten at a time.

Study the data displayed in Table 1 (below).

If we study the figures in the second column closely, we find some very interesting patterns.

Here are the sums arranged in a sequence (we have generated and included a few more terms):

18, 63, 165, 273, 408, 630, 828, 1053, 1395, 1683, 1998,
2460, 2838, 3243, 3825, 4293, 4788, 5490, 6048, 6633, ...

Selecting only the 1st, 4th, 7th, 10th, 13th, ... terms from above, we obtain these numbers:

18, 273, 828, 1683, 2838, 4293, 6048, ...

Multiples of 3	Sum of multiples of 3
1 to 10	18
1 to 20	63
1 to 30	165
1 to 40	273
1 to 50	408
1 to 60	630
1 to 70	828
1 to 80	1053
1 to 90	1395
1 to 100	1683
1 to 110	1998
1 to 120	2460

Table 1: Sums of multiples of 3 in sets of integers ten at a time

Let us compute the differences between consecutive numbers in the above sequence. Here is what we get:

$$255, 555, 855, 1155, 1455, 1755, \dots$$

A nice pattern!

Similarly, selecting only the 3rd, 6th, 9th, 12th, 15th, ... terms, we obtain these numbers:

$$165, 630, 1395, 2460, 3825, 5490, \dots$$

Just like we did earlier, let us compute the differences between consecutive numbers in the above sequence. Here is what we get:

465, 765, 1065, 1365, 1665, 1965, ...

Another nice pattern!

There are many more such patterns that can be uncovered. How do we explain them?

How do we explain the repeated occurrence of 300 in the above numbers?

We invite you to discover and study such patterns!

More explorations

Students can explore sums of multiples of other numbers (e.g., 4, 6 and 7) and find patterns in their consecutive sums.

After that we need to explain why these patterns exist!

About the author ...

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Playing with Primitive Pythagorean Triples – 1

It is possible to create interesting mathematics simply by playing with numbers and asking questions! Probably much of mathematics originated in exactly that way.

The idea of play is going to pervade most of the articles in this magazine.

The specific topic we shall study here is *Pythagorean triples*. As there is so much play possible with this topic, this is going to be a continuing series, running across many issues.

Keywords Coprime, Pythagorean triple, Primitive Pythagorean triple, PPT

Definitions

Coprime Positive integers a, b are said to be coprime if there is no divisor greater than 1 that divides both of a, b ; i.e., if $\text{GCD}(a, b) = 1$. Example: 4, 5 are coprime; 4, 6 are not coprime, as $\text{GCD}(4, 6) = 2 > 1$. Any two consecutive positive integers are coprime, and so are any two consecutive odd positive integers; do you see why?

Pythagorean triple A triple (a, b, c) of positive integers is said to be Pythagorean if $a^2 + b^2 = c^2$. Examples: (3, 4, 5) and (10, 24, 26)

Primitive Pythagorean triple (PPT) A Pythagorean triple (a, b, c) is called 'primitive' if there is no integer larger than 1 that divides all three of a, b, c , i.e., if $\text{GCD}(a, b, c) = 1$. For example, (3, 4, 5) is a PPT, but (10, 24, 26) is a Pythagorean triple that is not a PPT, as $\text{GCD}(10, 24, 26) = 2$.

A problem about Pythagorean triples

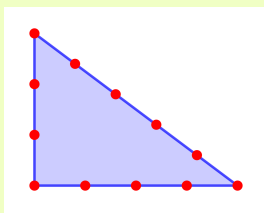


Figure 11: A triangle with sides 3, 4, 5

The triple (3, 4, 5) is surely one of the most familiar PPTs. Since

$$3^2 + 4^2 = 5^2, \quad (1)$$

we know that a triangle with sides 3, 4, 5 is right-angled.

There are many contexts where we (most unexpectedly!) meet a triangle with sides in the ratios 3 : 4 : 5. We will study this phenomenon later, in a separate article. For now, our focus is on another matter.

Note that the numbers 3, 4, 5 are consecutive. Seeing this, we embark on our first exploration:

Exploration – 1. Are there any other Pythagorean triples that consist of three consecutive numbers?

It is easy to show that the answer to this is No. This means that (3, 4, 5) is the only Pythagorean triple consisting of three consecutive numbers.

We had shown this in [1] but there is no harm in going over the proof again.

Proof

Let the sides of the triangle be $a - 1, a, a + 1$. Here, a is a positive integer, $a \geq 3$. (To see why $a \geq 3$, let's call upon the triangle inequality: "Any two sides of a triangle are together greater than the third side." It tells us that $(a - 1) + a > a + 1$, i.e., $a > 2$, which is the same as $a \geq 3$ since a is an integer.)

Since $(a - 1, a, a + 1)$ is a Pythagorean triple, we have

$$(a - 1)^2 + a^2 = (a + 1)^2,$$

$$\therefore 2a^2 - 2a + 1 = a^2 + 2a + 1,$$

$$\therefore a^2 = 4a,$$

$$\therefore a = 0 \text{ or } 4. \quad (2)$$

Since $a \geq 3$, it follows $a = 4$. Hence the triple is $(3, 4, 5)$. We have recovered our triple. ■

Extending the result – I

We may wonder: how about the shapes of triangles formed from other sets of three consecutive integers? It turns out that the following is true:

The triangle with sides 2, 3, 4 is obtuse-angled, and for $a > 4$, the triangles with sides $a - 1, a, a + 1$ are all acute-angled.

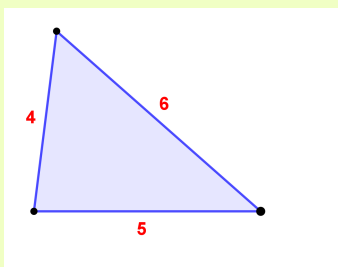
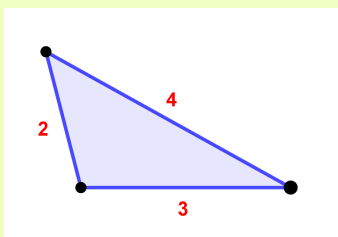


Figure 12: Triangles with sides 2, 3, 4 and 4, 5, 6

Try to prove this for yourself!

Extending the result – II

We can extend the above result in another direction altogether. Noting that the numbers 3, 4, 5 are in arithmetic progression (AP for short), we study the following problem:

Exploration – 2. Find all PPTs that consist of three positive integers in AP.

Solution

Let the sides of the triangle be $a - d, a, a + d$. Here, a and d are positive integers with $a > d$. The common difference of the AP is d .

To start with, we note that a and d must be coprime; for, if not, then there would be a factor greater than 1 dividing all three of $a - d, a, a + d$; i.e., we would have $\text{GCD}(a - d, a, a + d) > 1$. This would contradict the fact that we are dealing with a primitive Pythagorean triple.

Since $a - d, a, a + d$ are the sides of a right-angled triangle,

$$(a - d)^2 + a^2 = (a + d)^2,$$

$$\therefore 2a^2 - 2ad + d^2 = a^2 + 2ad + d^2,$$

$$\therefore a^2 = 4ad,$$

$$\therefore a = 4d. \quad (3)$$

Hence d is a divisor of a . But a, d are coprime; hence, $d = 1$. This yields $a = 4$, so the sides of the triangle are 3, 4, 5. We have thus obtained the same triple again. ■

More explorations

Let us relax the condition we had imposed in the first exploration. Instead of demanding that a, b, c are consecutive numbers, let us only insist that b and c are consecutive numbers. This little change allows us to generate infinitely many PPTs!

With the imposition of this new condition, let us consider the Pythagorean triple $(a, b, b+1)$. As the triple is Pythagorean we have

$$\begin{aligned}a^2 + b^2 &= (b+1)^2, \\ \therefore a^2 &= 2b+1.\end{aligned}\tag{4}$$

Since $2b+1$ is an odd number, this tells us that a is odd.

This immediately leads us to a simple way of generating as many Pythagorean triples as we wish:

- Choose any odd positive number $a > 1$.
- Find b and c using $b = \frac{1}{2}(a^2 - 1)$ and $c = b + 1$.
- Then (a, b, c) is a Pythagorean triple. Indeed it is a PPT since b, c , being consecutive integers, are coprime.

And that's it; as simple as that! So we get infinitely many PPTs using this simple recipe.

If $a = 1$, we obtain $(1, 0, 1)$. We do not classify this as a Pythagorean triple as it contains a 0. That is why we had insisted on $a > 1$. 

Using the above recipe we can generate infinitely many PPTs:

- $a = 3$ yields the PPT $(3, 4, 5)$.

- $a = 5$ yields the PPT $(5, 12, 13)$.
- $a = 7$ yields the PPT $(7, 24, 25)$.
- $a = 9$ yields the PPT $(9, 40, 41)$.

And so on.

Yet more explorations

Another promising exploration emerges from a small tweak: we now insist that b and c are consecutive odd numbers. Let us write the triple as $(a, b, b+2)$ and see where the algebra takes us. (Note that b must be odd.)

As the triple is Pythagorean we obtain:

$$\begin{aligned} a^2 + b^2 &= (b+2)^2, \\ \therefore a^2 &= 4b+4. \end{aligned} \tag{5}$$

The last relation tells us that a is even. Let $a = 2m$, where m is some positive integer. Then we get:

$$4m^2 = 4b+4, \quad \therefore b = m^2 - 1.$$

We had noted above that b must be odd; this tells us that m must be even. Let $m = 2n$, where n is some positive integer. We then have:

$$a = 4n, \quad b = 4n^2 - 1, \quad c = 4n^2 + 1,$$

So we have another infinite family of PPTs:

$$(4n, 4n^2 - 1, 4n^2 + 1). \tag{6}$$

This recipe too allows us to generate infinitely many PPTs:

- $n = 1$ yields the PPT $(4, 3, 5)$.
- $n = 2$ yields the PPT $(8, 15, 17)$.
- $n = 3$ yields the PPT $(12, 35, 37)$.
- $n = 4$ yields the PPT $(16, 63, 65)$.

And so on. The first item in the list might make you frown a little since the triple $(4, 3, 5)$ is just a permutation of the triple $(3, 4, 5)$. But we need not let this worry us.

As may be seen from the above, there are so many explorations possible in this simple environment; so many variations possible on this one theme. In the next issue, we shall continue these explorations still further.

Further reading

- [1] Community Mathematics Centre, "How many primitive Pythagorean triples in arithmetic progression?", *At Right Angles*, June 2012, pp. 33-34. <https://publications.azimpremjiuniversity.edu.in/1576/>

Article contributed by Shailesh Shirali

The Big Wide World of Data

Visualisation – 1

This column aims to present a running series of short articles on the theme of data visualisation. In recent years, with the increased focus on data, striking examples may be found of data depiction in imaginative ways. We shall present a few examples of this in each issue (with little or no commentary).

We start by sharing a beautifully written passage on the potentialities of imaginative visualisation, written by Henry Hubbard of the National Bureau of Standards, USA.

MAGIC IN GRAPHS

There is a magic in graphs. The profile of a curve reveals in a flash a whole situation – the life history of an epidemic, a panic, or an era of prosperity. The curve informs the mind, awakens the imagination, convinces.

Graphs carry the message home. A universal language, graphs convey information directly to the mind. Without complexity there is imaged to

the eye a magnitude to be remembered. Words have wings, but graphs interpret. Graphs are pure quantity, stripped of verbal sham, reduced to dimension, vivid, unescapable.

Graphs are all inclusive. No fact is too slight or too great to plot to a scale suited to the eye. Graphs may record the path of an ion or the orbit of the sun, the rise of a civilization, or the acceleration of a bullet, the climate of a century or the varying pressure of a heart beat, the growth of a business, or the nerve reactions of a child.

The graphic art depicts magnitudes to the eye. It does more. It compels the seeing of relations. We may portray by simple graphic methods whole masses of intricate routine, the organization of an enterprise, or the plan of a campaign. Graphs serve as storm signals for the manager, statesman, engineer; as potent narratives for the actuary, statist, naturalist; and as forceful engines of research for science, technology and industry. They display results. They disclose new facts and laws. They reveal discoveries as the bud unfolds the flower.

The graphic language is modern. We are learning its alphabet. That it will develop a lexicon and a literature marvelous for its vividness and the variety of application is inevitable.

Graphs are dynamic, dramatic. They may epitomize an epoch, each dot a fact, each slope an event, each curve a history. Wherever there are data to record, inferences to draw, or facts to tell, graphs furnish the unrivalled means whose power we are just beginning to realize and to apply.

Source: <https://www.maartenlambrechts.com/2017/01/25/another-code-to-charts-from-1939.html>

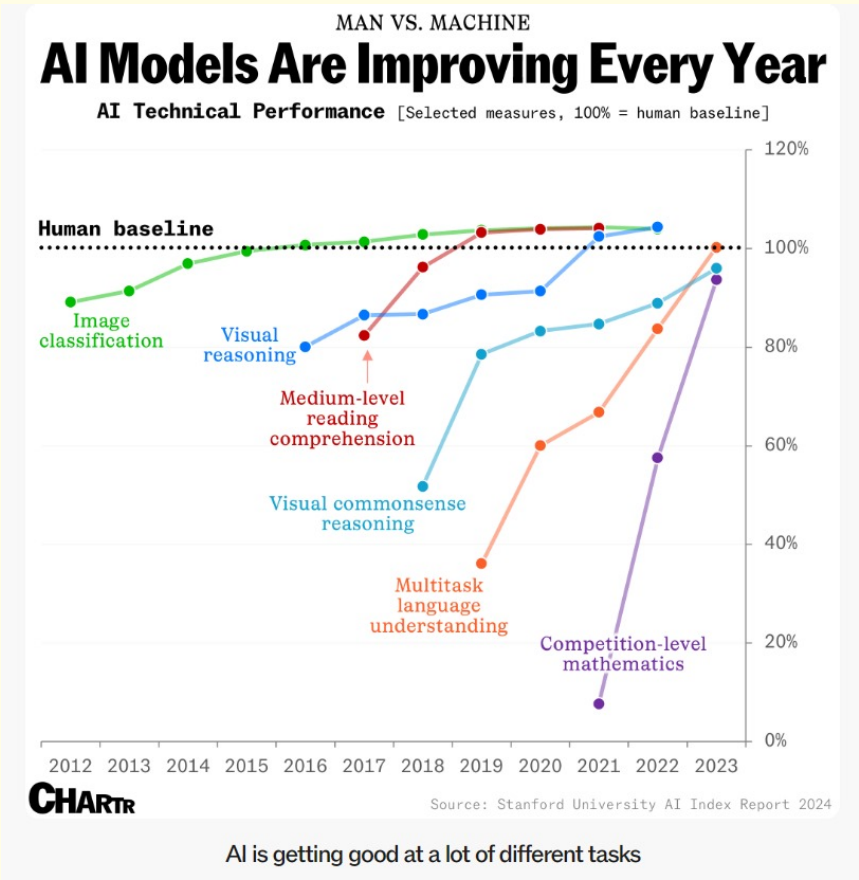


Figure 13: Credit: <https://sherwood.news/tech/how-do-ai-models-stack-up-vs-humans-on-standardized-benchmarks/>

World Youth. Over the next quarter-century, what is projected to happen to the distribution of young people around the world?

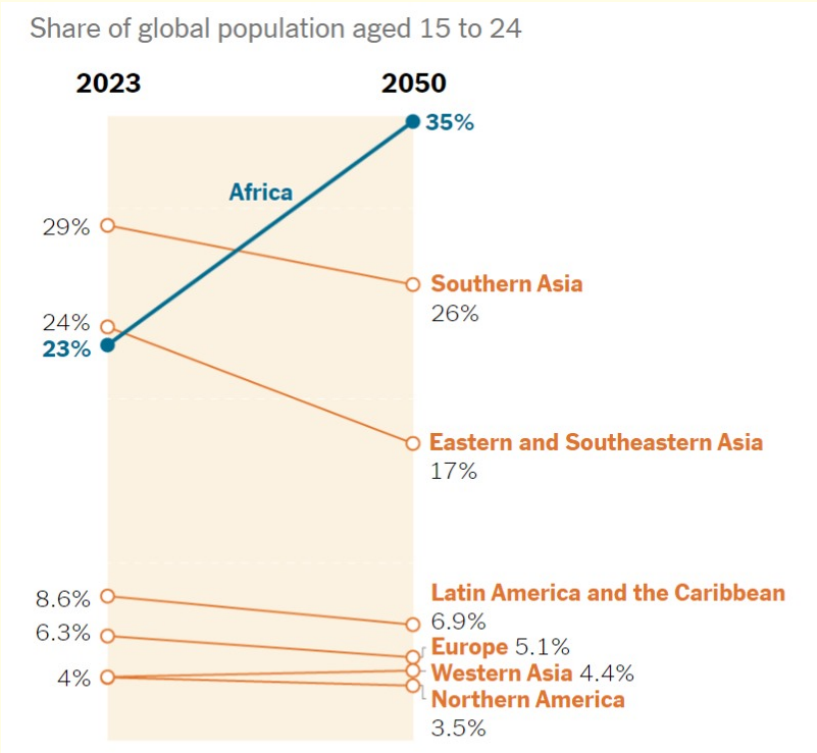


Figure 14: Source: Source: UN World Population Prospects 2022. Regions are based on UN classifications. Regions with less than 1% of the global population are not shown. By Lauren Leatherby. Credit: <https://www.nytimes.com/2024/03/14/learning/whats-going-on-in-this-graph-march-27-2024.html>

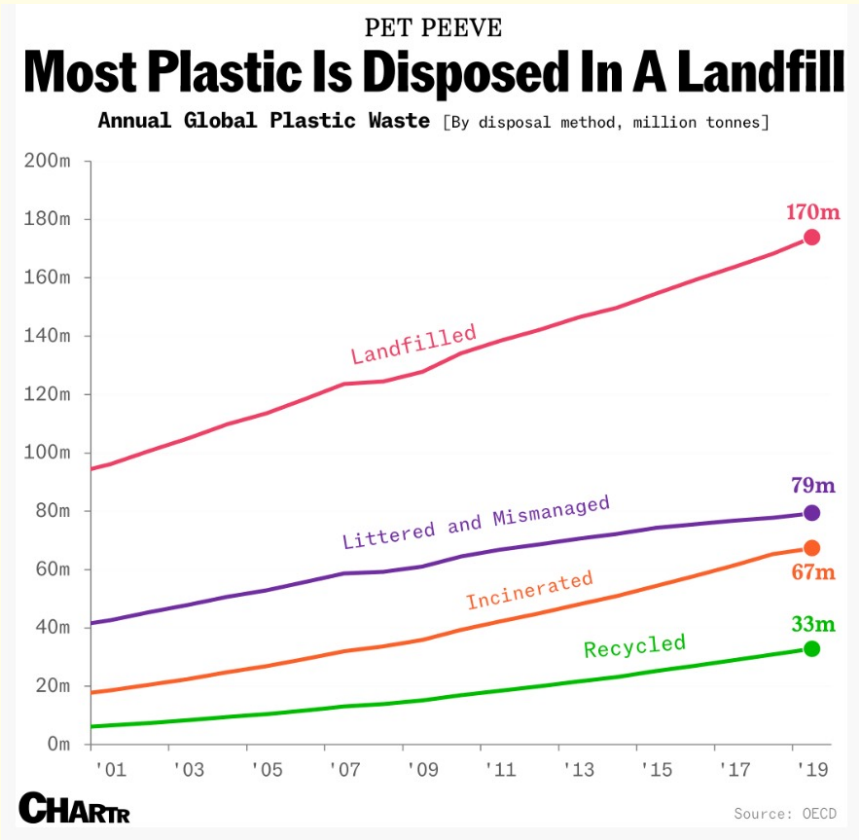


Figure 15: How much of the world’s plastic is recycled? Only a fraction, it seems. Source: <https://sherwood.news/world/how-much-of-the-worlds-plastic-is-recycled-only-a-fraction/>

Mathematicians Plan Computer Proof of Fermat's Last Theorem

Adapted from: <https://www.newscientist.com/article/2422601-mathematicians-plan-computer-proof-of-fermats-last-theorem/>. Author: Alex Wilkins

Fermat's Last Theorem (FLT for short) is the statement made by Pierre de Fermat in 1640 that there do not exist positive integers a, b, c satisfying the equation $a^n + b^n = c^n$ if n is any integer greater than 2. The statement is extremely intriguing, particularly as there exist infinitely many solutions in coprime positive integers to the Pythagorean equation $a^2 + b^2 = c^2$, and moreover, these solutions can be generated very easily.

Fermat had claimed to have a proof of the statement. This naturally adds to the intrigue but there is no evidence that he actually had a proof. The

simplicity of FLT's statement (together with Fermat's claim) invites us to try proving it — we get the feeling that there must be some simple one-page proof hiding beneath the surface. Over the centuries, large numbers of hopeful mathematicians battled with it, only to retire defeated. But many wonderful topics in mathematics emerged as a result of these attempts.

Then in 1993, Andrew Wiles of Princeton University announced a proof — not of the FLT itself but of another result called the *Modularity Conjecture*; this, together with results proved by many other mathematicians implies the FLT. But the proof was unbelievably complex; checking it for errors turned out to be very demanding.

(The story of FLT does not end there. A flaw was discovered in the proof, and it took a full year to fix it. So it would be more accurate to say that the theorem was proved in 1994-95, by Wiles along with his student Richard Taylor.)

Now, some mathematicians want to speed up this process by making proofs readable by computers. They plan to do this for Wiles's theorem using a programming language called **Lean**. The project, led by Kevin Buzzard and his team at Imperial College London, will follow Wiles's proof (with some changes). They plan to make their blueprint available online to enable contributions by others. The hope is that by formalizing the ideas behind Fermat's theorem, future research in number theory and arithmetic geometry will be the beneficiary.

Readers interested in reading about Lean could look up the following link:

<https://tinyurl.com/28so2tw>

Symmetry in the Natural World – 1

Here are some examples of five-fold symmetry from the natural world.

There is an important difference between the symmetry of the starfish and the periwinkle. See if you can spot it!

Remark

We referred to symmetry in our study of Heron's formula for the area of a triangle. We see that symmetry is a theme of relevance in mathematics as well as nature.



Figure 16: “Starfish resting on a sandy beach.” Photo by **Unsplash** from **Freerange Stock**, <https://freerangestock.com/photos/105578/starfish-resting-on-sandy-beach.html>



Figure 17: The Madagascar Periwinkle. Indian name: Nityakalyani. <https://www.pickpik.com/flower-madagascar-periwinkle-nityakalyani-indian-medicinal-plant-indian-medicinal-144203>

Mathematicians Discover ‘Soft Cell’ Shapes Behind the Natural World

Adapted from: <https://www.newscientist.com/article/2418791-mathematicians-discover-soft-cell-shapes-behind-the-natural-world/>. Author: Alex Wilkins

Mathematicians have found a new kind of shape called “soft cells” that match the curved shapes found in nature, like muscle cells and nautilus shells. This means we don’t have to only think about straight lines and sharp edges for now!

In the past, mathematicians focused on shapes like squares and triangles because they are easier to study mathematically. But these simple shapes don’t really represent the smooth curves we see in living things. Although models like polyhedra can describe natural objects like bubbles in foam,

they struggle with curved shapes like the cells that line our organs.

That's where soft cells come in. Gábor Domokos and his team created these shapes to have fewer sharp corners and fit together more closely. Muscle cells, which have two sharp corners, are a good example of a soft cell.

There are two types of soft cells: 2D ones with just two sharp points and many variations, and 3D ones with no sharp points at all, called "z-cells." Making these 3D shapes is still difficult, but researchers have found them in nature and even in architecture.

Architects already use shapes similar to soft cells to avoid corners, and z-cells appear in biological growth processes. The nautilus shell is a great example. These shells have a spiral structure, and when you cut through them, the chambers look like 2D soft cells. CT scans show that these chambers have no corners in 3D, suggesting that they are real-life z-cells.

This discovery could change how we describe natural structures. Soft cells offer a more realistic way to represent things than traditional shapes and could help us understand how the shape of soft tissues affects their properties. There are still many unanswered questions in this area.

How To Prove It – 1

This column will be a running series of articles on the theme of proof. Taking inspiration from the writings of the well-known mathematician George Pólya, we have named this column “How To Prove It” (after Pólya’s book *How To Solve It*). We will feature a variety of approaches used to prove or disprove statements.

In the first ‘episode’ we shall feature a well-known proof technique – proof by induction.

Proof by induction

Problems tackled using induction share some general features. The task is generally to prove a series of related assertions that hold for all positive integers n . For example, we may want to prove that

$$1^3 + 2^3 + 3^3 + \cdots + n^3 = (1 + 2 + 3 + \cdots + n)^2 \quad (7)$$

for all positive integers n . Note that this is actually a collection of infinitely many assertions (one for each value of n). Let us denote the above assertion

by Assertion(n). What we are being asked to prove is the infinite collection of assertions:

Assertion(1), Assertion(2), Assertion(3), Assertion(4), ...

It may seem that to prove infinitely many assertions is going to require an infinite amount of work! But we are able to get around this difficulty precisely because the assertions are linked to one another. If we can prove Assertion(1), and in addition we can prove that if any one assertion in the collection is true, then so is the next one, then we will have proved the following infinite chain:

$$\text{Assertion}(1) \implies \text{Assertion}(2) \implies \text{Assertion}(3) \implies \dots \quad (8)$$

As we have already proved Assertion(1), it means that in principle we have proved all the statements!

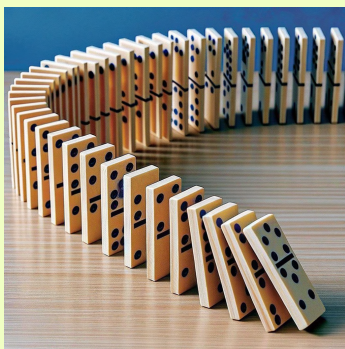


Figure 18: The Domino Effect: each domino knocks over the next one

A proof organised along these lines is called a *proof by induction*. Perhaps we may want to call it the *Domino Effect*! (See Figure 18.)

Illustrative problems for study

We list a few illustrative examples below. We have deliberately chosen problems that require a slight twist in the strategy each time. Please study these examples carefully.

1. Prove that $x - 1$ is a factor of $x^n - 1$ for all positive integers n .
2. Prove that $2^n + 1$ is a multiple of 3 for all *odd* positive integers n .
3. Prove that $1^n + 2^n + 3^n + 4^n$ is a multiple of 5 precisely when n is *not* a multiple of 4.
4. Let f_n denote the n -th Fibonacci number. The Fibonacci numbers are 1, 1, 2, 3, 5, 8, 13, 21, ..., so we have $f_1 = 1, f_2 = 1, f_3 = 2, f_4 = 3, f_5 = 5$, and so on. Prove that every 5th Fibonacci number is a multiple of 5 (i.e., $f_5, f_{10}, f_{15}, \dots$ are multiples of 5).

Solution of problem 1

We must look for a way of connecting $x^n - 1$ with $x^{n+1} - 1$, and do it in such a way that the role played by $x - 1$ becomes clear. But this is accomplished by a simple subtraction:

$$\begin{aligned}(x^{n+1} - 1) - (x^n - 1) &= x^{n+1} - x^n \\ &= x^n(x - 1) = \text{a multiple of } (x - 1).\end{aligned}$$

That is:

$$(x^{n+1} - 1) = (x^n - 1) + \text{a multiple of } (x - 1). \quad (9)$$

It follows from (9) that if $x - 1$ is a factor of $x^n - 1$, then $x - 1$ is a factor of $x^{n+1} - 1$ as well. We have found the desired connection.

So, denoting the claim that $x - 1$ is a factor of $x^n - 1$ by $\text{Assertion}(n)$, we have shown the following:

$$\text{Assertion}(n) \implies \text{Assertion}(n+1),$$

and we have thus established the infinite chain:

$$\text{Assertion}(1) \implies \text{Assertion}(2) \implies \text{Assertion}(3) \implies \dots$$

Since $\text{Assertion}(1)$ is true, it follows that $\text{Assertion}(n)$ is true for all positive integers n . ■

Solution of problem 2

The claim that $2^n + 1$ is divisible by 3 appears to be false for all even values of n . For example, $2^2 + 1 = 5$ is not divisible by 3; nor is $2^4 + 1 = 17$. But the claim appears to be true for odd values of n ; for example, $2^1 + 1 = 3$ is divisible by 3; so is $2^3 + 1 = 9$.

This means that we must look for a way of connecting $2^n + 1$ with $2^{n+2} + 1$; for if n is an odd number, then the next odd number is $n + 2$ (and not $n + 1$).

This is yet again accomplished by simple subtraction; we have:

$$\begin{aligned}(2^{n+2} + 1) - (2^n + 1) &= 2^{n+2} - 2^n \\ &= 2^n \times (2^2 - 1) = 2^n \times 3.\end{aligned}$$

So the difference between the two quantities is a multiple of 3. It follows that if $2^n + 1$ is a multiple of 3, then $2^{n+2} + 1$ too is a multiple of 3. We have found the desired connection.

Since $2^1 + 1$ is a multiple of 3, it follows that all of the quantities $2^3 + 1$, $2^5 + 1$, $2^7 + 1$, ... are multiples of 3. We have therefore proved that $2^n + 1$ is a multiple of 3 for all odd positive integers n . ■

Solution of problem 3

Let $a(n) = 1^n + 2^n + 3^n + 4^n$. We tabulate the values of $a(n)$ for the first few values of n :

n	1	2	3	4	5	6	7	8	9	...
$a(n)$	10	30	100	354	1300	4890	18700	72354	282340	...

We spot a striking pattern immediately: $a(n)$ has units digit 0 for all values of n except when n is a multiple of 4, and when that happens the units digit is 4. We now have a good working hypothesis and we can set about proving it using induction.

The observation that ‘something noteworthy’ is happening at multiples of 4 suggests that we should look for a connection between $a(n)$ and $a(n+4)$. Just as we did earlier, let us see where simple subtraction leads us:

$$\begin{aligned}
 a(n+4) - a(n) &= (1^{n+4} + 2^{n+4} + 3^{n+4} + 4^{n+4}) - (1^n + 2^n + 3^n + 4^n) \\
 &= 2^n \cdot (2^4 - 1) + 3^n \cdot (3^4 - 1) + 4^n \cdot (4^4 - 1) \\
 &= 2^n \cdot 15 + 3^n \cdot 80 + 4^n \cdot 255.
 \end{aligned}$$

Since the individual terms $2^n \times 15$, $3^n \times 80$ and $4^n \times 255$ are multiples of 10, it follows that $a(n+4) - a(n)$ is a multiple of 10 for all positive integers n . This means that $a(n)$ and $a(n+4)$ have the same units digit, for all n .

Since $a(1), a(2), a(3), a(4)$ have units digits 0, 0, 0, 4, it follows that the cycle 0, 0, 0, 4 repeats every 4 steps.

The required conclusion follows: that the quantity $1^n + 2^n + 3^n + 4^n$ is a multiple of 5 precisely when n is not divisible by 4. ■

Solution of problem 4

The Fibonacci numbers are defined additively: each number after the first two is the sum of the preceding two numbers. Let us see what happens to the remainders when these numbers are divided by 5. Naturally, the remainders follow the same additive law (with the understanding that addition happens in the “modulo 5 world” with $1 + 4 = 0$, $2 + 4 = 1$, and so on). We obtain the following numbers:

$$\underbrace{1, 1, 2, 3, 0, 3, 3, 1, 4, 0, 4, 4, 3, 2, 0, 2, 2, 4, 1, 0}_{\text{Cycle of length 20}}, 1, 1, 2, 3, \dots$$

After 20 steps we obtain the remainders 1, 1. That is, the 21st and 22nd terms in the remainder sequence are the same as the 1st and 2nd terms. It follows that the remainders cycle after every 20 steps. We also observe that in the repeating portion, the 0s occur at positions 5, 10, 15 and 20. The required conclusion thus follows: that every 5th Fibonacci number is a multiple of 5. ■

Comment

We have not written this proof in the usual “proof-by-induction” style. But if you observe carefully, you will find the inductive reasoning hidden in the above sentences.

For a more direct proof we proceed as follows. For any given value of n , let $f_n = a$ and $f_{n+1} = b$. The succeeding Fibonacci numbers are then:

$$f_{n+2} = a + b, \quad f_{n+3} = a + 2b, \quad f_{n+4} = 2a + 3b, \quad f_{n+5} = 3a + 5b.$$

So

$$f_{n+5} = 3f_n + 5f_{n+1}.$$

It follows that if f_n is a multiple of 5, then so is f_{n+5} . Since $f_5 = 5$, it follows that every 5th Fibonacci number is a multiple of 5. ■

We shall study more such problems in future articles of this column.

How a Rare Islamic Astrolabe Helped Muslims, Jews and Christians Tell Time and Read Horoscopes

Adapted from: <https://www.scientificamerican.com/article/islamic-astrolabe-helped-muslims-jews-and-christians-tell-time/>. Author: Annie Melchor

A Multilingual Stargazer

A long-lost device called an astrolabe was found in Verona, Italy. It tells an interesting story about science, travel, and different cultures. This small brass disc, made in 11th-century Spain when it was ruled by Muslims, has travelled across many centuries and continents.



Figure 19: The Verona astrolabe. Credit: Federica Gigante

The astrolabe was like a portable computer for studying the stars. It helped people tell time, find directions, and even read horoscopes. It had many moving parts that showed the positions of stars in the sky. By aligning the star map with the user's location and the date, the astrolabe could tell the time and give other details about the stars.

What makes this astrolabe special is its markings in different languages. It was originally written in Arabic, but there are also Hebrew and Western numbers on it. These changes suggest that different people used the astrolabe, including Muslims, Jews, and likely Christian Italians. They modified it to suit their own needs. For example, Hebrew translations

were added for those who didn't understand Arabic, and changes to the latitude markings showed that it was used in different places.

Historians think the astrolabe first came from Al-Andalus (Muslim Spain) and then travelled to Italy. The Hebrew markings suggest that it was owned by Jewish communities. Eventually, it ended up in the collection of an Italian nobleman in the 17th century, probably between 1656 and 1672. Today, it is kept in a museum in Verona. Besides its scientific uses, the astrolabe also had religious meaning. Muslims used it to figure out prayer times and the direction of Mecca. It was also used in astrology, which was important in many religions at that time. Astrology helped people make decisions about their lives, like social, economic, and medical choices.

However, during the Scientific Revolution, people started focusing more on science and less on astrology. New instruments were developed for studying the stars, and the astrolabe became less important.

The rediscovered Verona astrolabe, with its layers of history and different languages, shows us how people in the past were connected through science.

It also reminds us that astronomy and astrology are age-old pursuits of mankind. We have always been fascinated by the stars and by our place in the universe.

Solar Eclipse of 8th April, 2024

The total solar eclipse of 8 April 2024 which was visible to large numbers of people in Mexico and the United States aroused considerable interest. The reader would surely have come across lyrical write-ups about the eclipse in various publications, with witnesses describing it as an emotional and unforgettable experience.

We came across the following passage in the online magazine *Live Science*:

With the April 8 total solar eclipse over, North America has exactly 8 years, 11 months and 22 days to wait until its next one. Although total solar eclipses occur in the same location only every 375 years or so, they are not rare. In fact, they happen on our planet about once every 18 months, on average.

In the next decade, seven total solar eclipses will occur on Earth, plunging countries as diverse as Australia, Egypt, Spain and Sudan under the central shadow of the moon. The first of



Figure 20: Image of the total solar eclipse of April 8, 2024, in Arkansas, USA. Taken from: <https://tinyurl.com/25tmmpgh>

these will be on Aug. 12, 2026, and the last will be on March 20, 2034, with maximum durations of totality ranging from 1 minute, 8 seconds to 6 minutes, 23 seconds. (Source: <https://tinyurl.com/2mtudjym>)

We also came across the following statement in Wikipedia:

The longest possible duration of a total solar eclipse is 7 minutes and 32 seconds. The longest annular solar eclipse of the 21st century took place on January 15, 2010, with a duration of 11 minutes and 7.8 seconds. The maximum possible duration is 12 minutes and 29 seconds. (Source: https://en.wikipedia.org/wiki/List_of_solar_eclipses_in_the_21st_century)

There are several remarkable statements made in these two passages, and they will surely excite our curiosity. For example:

- “Total solar eclipses occur in the same location only every 375 years or so.”

On what basis can we make such a statement? From where do we obtain the figure of 375 years?

- “[Total solar eclipses] happen on our planet about once every 18 months, on average.”

On what basis can we make such a statement?

- Claims have been made about the duration of totality of various solar eclipses. How do we compute (in advance) the duration of totality of a total solar eclipse?

- “The longest possible duration of a total solar eclipse is 7 minutes and 32 seconds. . . . The maximum possible duration [of an annular solar eclipse] is 12 minutes and 29 seconds.”

On what basis can we make all these remarkable statements? How do we estimate the maximum possible durations of a total solar eclipse and an annular solar eclipse with such precision?

It should be clear that there is plenty of deep mathematics (geometry and algebra and trigonometry and physics) behind a solar eclipse!

For now, we will leave you with these questions. In the next issue of this magazine, we will try to justify the statements made above.

The Earth Is Slowing Down in Its Rotation

Adapted from: <https://www.scientificamerican.com/article/global-warming-is-slowing-the-earths-rotation/>. Author: Avery Schuyler Nunn

A new study has found that the melting of ice at the North and South Poles is causing the Earth to spin more slowly. This happens because when the ice melts it moves, some of the Earth's mass away from the North Pole and thus away from the axis of the Earth. This slows down the Earth's rotation. It's like when figure skaters extend their arms, they slow down. The Earth's core is also speeding up a little bit (for different reasons), but overall, the slowing-down effect is stronger.

Right now, we use something called "leap seconds" to keep our clocks in sync with the Earth's rotation. Leap seconds are added or subtracted from the time to make up for the changes in the Earth's rotation. But

because the ice melt is slowing down the Earth's rotation and hiding the core's speeding up, we might have to wait longer before we subtract a leap second.

According to the study, we might need to subtract a leap second around 2028 or 2029, but we're not exactly sure. This would be the first time we ever subtract a leap second, and it could cause problems for computer networks and GPS devices that rely on precise timing.

Some scientists think that leap seconds are unnecessary because the Earth's rotation is always changing, and they want to get rid of them by 2035. This new research makes the debate about how to handle the difference between the Earth's rotation and our need for accurate timekeeping more complicated.

You've Heard of Dyslexia. But Why Not Dyscalculia?

Adapted from: <https://www.scientificamerican.com/article/understanding-dyscalculia-dyslexias-numeric-counterpart/>. Author: Jaimie Seaton

This article talks about a problem called *developmental dyscalculia*, which is a learning disorder that affects how people understand math and numbers. It is estimated to affect 3 to 7 percent of people worldwide, but it often goes undiagnosed and not many people know about it. It can cause difficulties with things like understanding patterns, managing time, knowing where things are in space, and following instructions in order. This makes everyday activities harder for people with dyscalculia.

Repercussions of Dyscalculia: Dyscalculia is also connected to other mental health and brain conditions like anxiety, depression, ADHD, sensory processing disorder, autism spectrum disorder, and dyslexia. But dyslexia

is much better recognized and understood in comparison with dyscalculia. Children with dyslexia are 100 times more likely to get diagnosed and receive help than children with dyscalculia.

Challenges in Understanding Dyscalculia: Understanding dyscalculia is challenging because there isn't a clear cause or set of symptoms. Different research studies have conflicting findings, and there are no standardized rules for diagnosing it. The debate is whether dyscalculia comes from a problem with understanding numbers or not being able to see them properly.

Researchers say it's important to understand the thinking skills needed for math, like memory and problem-solving. But it's hard to separate these skills from the specific math skills, so it's tough to figure out what exactly causes dyscalculia. Also, things like the environment and situation can affect dyscalculia, so there are many different things to think about when studying it.

Ongoing Effort to Understand Dyscalculia: The article talks about ongoing efforts to understand dyscalculia better. It says we need a clear definition of dyscalculia and more research with bigger groups of people. It also suggests that researchers from different places need to work together to have the same rules for studying dyscalculia.

It is important to raise awareness about these matters, particularly within the teacher community. We certainly need a better understanding of the phenomenon of dyscalculia.

Two Solutions to a Geometry Problem

Article contributed by Anudeep Prashanth & Dhruv Perody

In this article, we discuss two solutions to a geometry problem from Round 2 of the Hungarian Mathematical Olympiad 2001/02. The problem was given to us during the weekend problem solving group of our school. We find that both the solutions involve ideas from classical geometry as well as trigonometry, but the approaches are different.

Keywords Cyclic quadrilateral, congruent triangles, similar triangles, trigonometric ratio, sine rule, compound angle formula for sine

Problem

The perpendicular from vertex A of the rectangle $ABCD$ to diagonal BD meets BD and side BC at the interior points E and F , respectively. Determine the ratio of the adjacent sides of $ABCD$ if segment EF subtends an

angle of 30° at C .

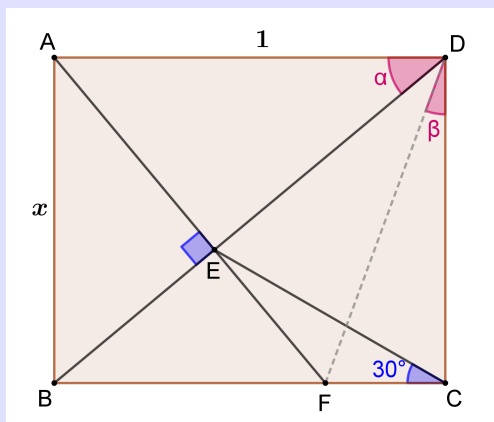


Figure 21

Solution 1

Let $\angle BDA = \alpha$ and $\angle CDF = \beta$ (see Figure 21). Observe that $EFCD$ is a cyclic quadrilateral and note that $\angle EDF = \angle ECF = 30^\circ$. Hence we have $\alpha + \beta = 60^\circ$. This gives

$$\sin \alpha \cos \beta + \sin \beta \cos \alpha = \frac{\sqrt{3}}{2}. \quad (10)$$

We now take $AD = 1$ and $AB = x$, express various trigonometric ratios in terms of x , and then solve for x .

For this, observe by AAA similarity that $\triangle ABF \sim \triangle BCD$. This implies

$$\frac{BF}{CD} = \frac{AB}{BC},$$

$$\therefore BF \cdot BC = AB \cdot CD,$$

$$\therefore BF = \frac{AB \cdot CD}{BC} = \frac{x \cdot x}{1},$$

giving $BF = x^2$, and so:

$$\sin \alpha = \frac{x}{\sqrt{1+x^2}}, \quad \cos \alpha = \frac{1}{\sqrt{1+x^2}}.$$

Also, $FC = 1 - x^2$, so $FD^2 = x^2 + (1 - x^2)^2 = x^4 - x^2 + 1$. Therefore:

$$\sin \beta = \frac{1 - x^2}{\sqrt{x^4 - x^2 + 1}}, \quad \cos \beta = \frac{x}{\sqrt{x^4 - x^2 + 1}}.$$

From equation (10) we obtain:

$$\frac{x}{\sqrt{1+x^2}} \cdot \frac{x}{\sqrt{x^4 - x^2 + 1}} + \frac{1 - x^2}{\sqrt{x^4 - x^2 + 1}} \cdot \frac{1}{\sqrt{1+x^2}} = \frac{\sqrt{3}}{2},$$

i.e.,

$$\frac{1}{\sqrt{1+x^2} \cdot \sqrt{x^4 - x^2 + 1}} = \frac{\sqrt{3}}{2}.$$

This simplifies to

$$(x^4 - x^2 + 1) \cdot (x^2 + 1) = \frac{4}{3},$$

and then to

$$x^6 = \frac{1}{3}. \quad (11)$$

The only real positive root of this equation is $x = 1/3^{1/6}$.

So $AB : AD = 1/3^{1/6}$. ■

Solution 2

As earlier, let $AD = 1$, $AB = x$ (see Figure 22).

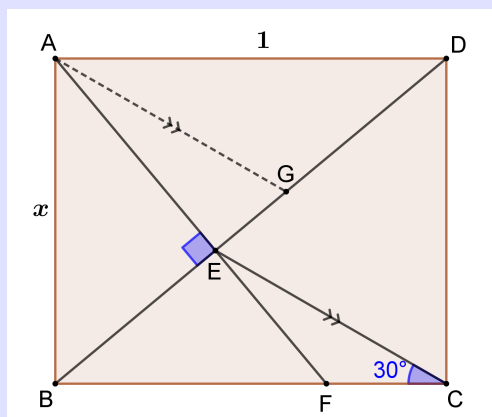


Figure 22

Note that $\triangle EBA$ is similar to $\triangle ABD$. Thus,

$$\frac{EB}{AB} = \frac{AB}{BD}, \quad \therefore BE = \frac{AB^2}{BD} = \frac{x^2}{\sqrt{1+x^2}}. \quad (12)$$

Also,

$$\frac{AE}{AD} = \frac{AB}{BD}, \quad \therefore AE = \frac{x}{\sqrt{1+x^2}}. \quad (13)$$

Let G be an interior point of BD such that AG is parallel to CE . Then $\triangle AGD$ is congruent to $\triangle CEB$, so $\angle GAD = 30^\circ$ and $GD = BE$.

Using $\angle AGE = 180^\circ - \angle AGD$ (which means that $\sin AGE = \sin AGD$) and the sine rule in $\triangle AGD$, we find:

$$\sin AGE = \frac{\sin GAD}{GD} \cdot AD = \frac{\sin 30^\circ}{BE} = \frac{\sqrt{1+x^2}}{2x^2}.$$

Therefore,

$$\tan AGE = \frac{\sin AGE}{\sqrt{1 - \sin^2 AGE}} = \frac{\sqrt{1+x^2}}{\sqrt{4x^4 - 1 - x^2}}, \quad (14)$$

after simplification. From the expression for AE and

$$GE = BD - 2BE = \sqrt{1+x^2} - \frac{2x^2}{\sqrt{1+x^2}} = \frac{1-x^2}{\sqrt{1+x^2}},$$

we find, in the right-angled triangle AGE ,

$$\tan AGE = \frac{AE}{GE} = \frac{x}{1-x^2}. \quad (15)$$

Hence we get

$$\frac{\sqrt{1+x^2}}{\sqrt{4x^4 - 1 - x^2}} = \frac{x}{1-x^2}.$$

From this, we get

$$(1+x^2)(1-x^2)^2 = x^2(4x^4 - 1 - x^2),$$

$$\therefore 3x^6 = 1 \quad (\text{on simplification})$$

Thus we get, as earlier:

$$x = \frac{1}{3^{1/6}}.$$



Acknowledgement We wish to thank our Math Teacher, Shri P N Subrahmanya, for guiding us in writing this article.

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Further reading

[1] *Problem collection for Hungary*, <https://imomath.com>

An Area Problem

In this short article we consider a problem from geometry: finding the area of a rectangle in a simple configuration. What makes the problem attractive is that it has multiple solutions, each instructive and elegant in its own way. The problem was posed by Shri C Mahipal Reddy of Telangana.

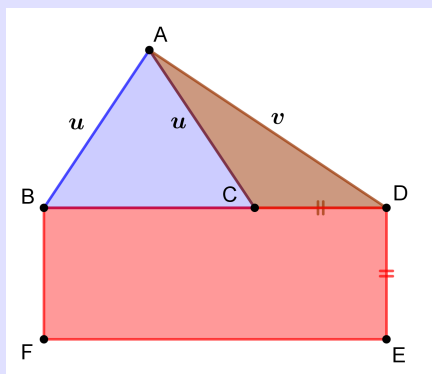


Figure 23: $AB = AC$; $CD = DE$; to find the area of rectangle $BDEF$ in terms of u, v

Problem statement

Figure 23 shows an isosceles triangle ABC (with $AB = AC$) and triangle ACD ; the two triangles share side AC ; points B, C, D are collinear. A rectangle is drawn with side $DE = CD$. It is given that $AB = u = AC$ and $AD = v$. The problem is to find the area of rectangle $BDEF$ in terms of u and v .

Solution 1

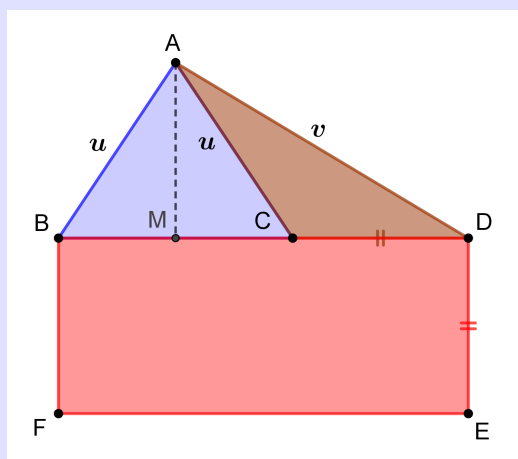


Figure 24

The area is $DE \cdot DB = DC \cdot DB$, since $DE = DC$. Join A to the midpoint M of BC (Figure 24).

Then $AM \perp BC$, as $\triangle ABC$ is isosceles.

We now have:

$$\begin{aligned}
 DC \cdot DB &= (DM - CM) \cdot (DM + MB) \\
 &= (DM - CM) \cdot (DM + CM) = DM^2 - CM^2 \\
 &= (DM^2 + AM^2) - (CM^2 + AM^2) \\
 &= DA^2 - CA^2 = v^2 - u^2.
 \end{aligned}$$

So the required area is $v^2 - u^2$. ■

Solution 2 (due to Shri Durgesh Bannu)

This is almost a proof without words!

As noted at the start of the earlier proof, the area of the rectangle is equal to $DC \cdot DB$.

Draw a circle centered at A and passing through B (and, therefore, C too). Its radius is u . (See Figure 25.)

Draw a tangent from D to the circle. Let the tangent touch the circle at H . Then triangle ADH is right-angled at H . Hence $DH = \sqrt{v^2 - u^2}$.

By the intersecting chords theorem (see Figure 26 on page 77 for what this theorem asserts) applied to the case of tangency,

$$\begin{aligned}
 DC \cdot DB &= DH^2, \\
 \therefore DC \cdot DB &= v^2 - u^2.
 \end{aligned}$$

So the area of the rectangle is $v^2 - u^2$. ■

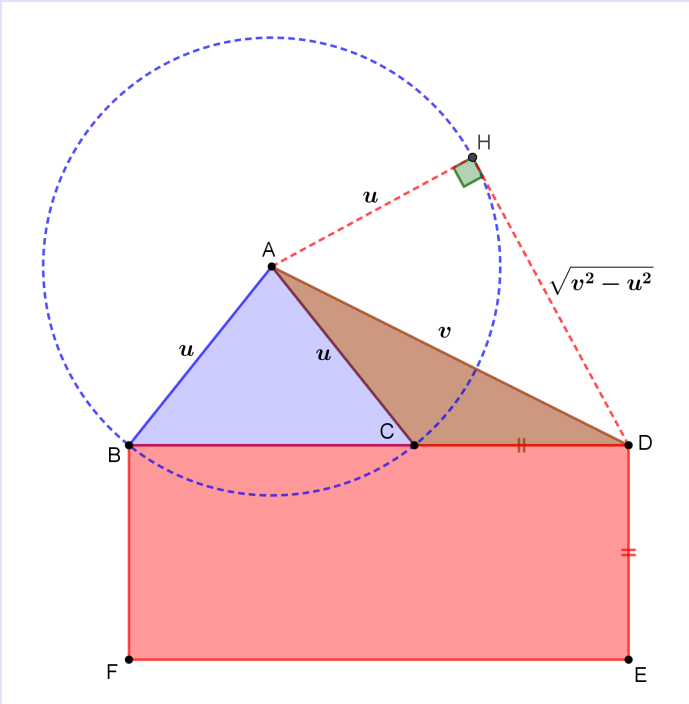


Figure 25

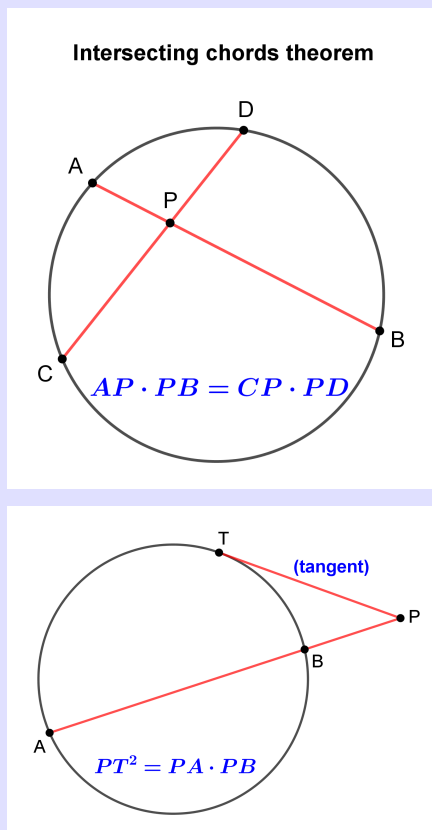


Figure 26: Intersecting chords theorem

A Tale of Two Sieves

In the kitchen we use a sieve to filter out something unwanted, or to let through something we want, from a mixture of different kinds of objects. A sieve in number theory plays the same role: it is an algorithm used to collect numbers of some desired kind, or to filter them out, from a larger set.

In this article we describe two sieves that filter out the primes from the natural numbers. The first sieve is well-known; it is studied at the upper primary level. (We describe it below, for the sake of completeness.) The second sieve is hardly known at all and we are happy to bring it to your attention.

The sieve of Eratosthenes

Eratosthenes was a Greek scholar who lived in the 3rd century BCE. He was an expert in so many fields (mathematics, astronomy, geography, poetry) that he was nicknamed “Beta of Cyrene” — as being the second best, after Plato! (It must have been quite a challenge working alongside Plato!) In

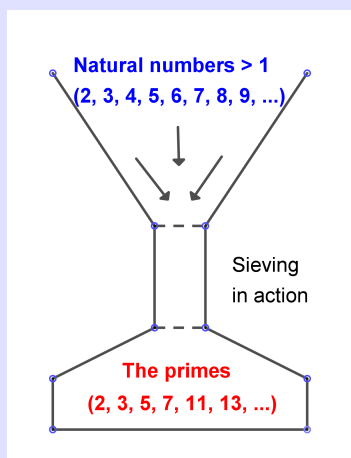


Figure 27: The sieve of Eratosthenes in action

mathematics, he devised an algorithm for generating prime numbers; it is still used today. In astronomy, he worked out a way of estimating the circumference of the Earth using the noon sun's shadow at two locations far apart. He also created a star catalogue that was in use centuries later. He was the head librarian at the Alexandria Library (at Alexandria, in present day Egypt), a very famous centre of knowledge in ancient times. We cannot really call him a Beta!

How the Sieve of Eratosthenes works

The sieve finds all primes up to a given limit N . It works as follows. Keep in mind that the number 1 is neither prime nor composite; it is a 'unit' of the natural number system.

Step 0 Create a list $\mathcal{L}$ of numbers from 2 to N . Initially, all the numbers are **unmarked**. Later, the numbers that get **marked** are the primes, and the numbers that get deleted are the composite numbers.

Step 1 Scan the updated list $\mathcal{L}$ and identify the **smallest unmarked number**; call it k . Mark k and designate it as a prime number.

Step 2 From $\mathcal{L}$, **delete all multiples of k** (the number identified in Step 1) other than k itself. That is, delete the numbers $2k, 3k, 4k, \dots$

Step 3 Repeat Step 1 and Step 2 (in a cycle) till every number in $\mathcal{L}$ is marked. When that stage is reached, the algorithm terminates. *The marked numbers are the prime numbers less than or equal to N .*

For example, when applying the sieve to find all the primes less than or equal to $N = 30$, we obtain the successive lists shown in Table 1. (For visual clarity, we have shown the marked numbers enclosed in boxes.)

It should be clear why this sieve works. On the first run through, it filters out all multiples of 2, except 2 itself. Then, in the next round, it filters out all multiples of 3, except 3 itself. Following this, it filters out all multiples of 5, except 5 itself. And so it proceeds, steadily filtering out all the composite numbers. What is left in the end must clearly be the set of prime numbers.

Python code for the Sieve of Eratosthenes

Here is the python code for putting the sieve in action.

Source: <https://www.geeksforgeeks.org/python-program-for-sieve-of-eratosthenes/>

```
# Python program to print all Primes Smaller
# than or equal to N using Sieve of Eratosthenes
```

Step	Updated $\mathcal{L}$	Updated $\mathcal{P}$
0	2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30	
1	$\boxed{2}$, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30	2
2	$\boxed{2}$, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29	2
1	$\boxed{2}$, $\boxed{3}$, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29	2, 3
2	$\boxed{2}$, $\boxed{3}$, 5, 7, 11, 13, 17, 19, 23, 29	2, 3
1	$\boxed{2}$, $\boxed{3}$, $\boxed{5}$, 7, 11, 13, 17, 19, 23, 29	2, 3, 5

Table 2: First few cycles of the Sieve of Eratosthenes

```
def SieveOfEratosthenes (num) :  
    prime = [True for i in range(num+1)]  
    # boolean array  
    p = 2  
    while (p * p <= num) :  
  
        # If prime[p] is not  
        # changed, then it is a prime
```

```
if (prime[p] == True):

    # Updating all multiples of p
    for i in range(p * p, num+1, p):
        prime[i] = False
    p += 1

# Print all prime numbers
for p in range(2, num+1):
    if prime[p]:
        print(p)

# Driver code
if __name__ == '__main__':
    num = 40
    print("Following are the prime numbers smaller"),
    print("than or equal to", num)
    SieveOfEratosthenes(num)
```

Sample output

On running the above code we get the following output:

```
Following are the prime numbers smaller
than or equal to 50
2
3
5
7
```

11

13

17

19

23

29

31

37

41

43

47

```
=== Code Execution Successful ===
```

Sundaram's sieve

This sieve was discovered by an Indian mathematics student S. P. Sundaram in 1934 and initially introduced as a simpler alternative to the Sieve of Eratosthenes.

The algorithm works by constructing the following symmetric array $\mathcal{A}$ of numbers in which each row and each column is an infinite arithmetic

progression (AP):

$$\mathcal{A} = \begin{array}{ccccccc} 4 & 7 & 10 & 13 & 16 & 19 & \dots \\ 7 & 12 & 17 & 22 & 27 & 32 & \dots \\ 10 & 17 & 24 & 31 & 38 & 45 & \dots \\ 13 & 22 & 31 & 40 & 49 & 58 & \dots \\ 16 & 27 & 38 & 49 & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \end{array}$$

Note its structure:

- The 1st row has the AP 4, 7, 10, 13, ... with first term 4 and common difference 3. Same for the 1st column.
- The 2nd row has the AP with first term 7 and common difference 5. Same for the 2nd column.
- The 3rd row has the AP with first term 10 and common difference 7. Same for the 3rd column.
- The 4th row has the AP with first term 13 and common difference 9. Same for the 4th column.

And so on.

Sundaram's rule is the following. Let L_A be the list of numbers occurring in $\mathcal{A}$. Produce a second list L_B of numbers by doubling all the numbers in L_A and adding 1.

So, since

$$L_A = \{4, 7, 10, 12, 13, 16, 17, 19, \dots\},$$

the numbers in the second list are

$$L_B = \{9, 15, 21, 25, 27, 33, 35, 39, \dots\}.$$

It happens that the numbers in L_B are just the odd composite numbers, so an odd number greater than 1 is prime if and only if it does not occur in L_B .

This may seem mystifying at first sight but it is easy to understand. All we need to do is to produce a second array $\mathcal{B}$ from $\mathcal{A}$ by doubling the numbers in $\mathcal{A}$ and adding 1. Here is what we get:

$\mathcal{B} =$	9	15	21	27	33	39	...
	15	25	35	45	55	65	...
	21	35	49	63	77	91	...
	27	45	63	81	99	117	...
	33	55	77	99	...	...	...
	...	...	...	...	...	...	...

We observe that:

- The 1st row has all the odd multiples of 3, starting with 3×3 .
- The 2nd row has all the odd multiples of 5, starting with 5×3 .
- The 3rd row has all the odd multiples of 7, starting with 7×3 .
- The 4th row has all the odd multiples of 9, starting with 9×3 .

And so on. This means that $\mathcal{B}$ is composed entirely of odd composite numbers, just as we had noted.

To understand this algebraically, note that in Sundaram's original array $\mathcal{A}$, the number in the 1st row and column j is $4 + 3(j - 1)$; the number in the 2nd row and column j is $7 + 5(j - 1)$; the number in the 3rd row and column j is $10 + 7(j - 1)$; and so on. In general, the number in row i and column j in $\mathcal{A}$ is

$$a(i, j) = 4 + 3(i - 1) + (2i + 1)(j - 1) = 2ij + i + j.$$

Hence, the number in row i and column j in $\mathcal{B}$ is

$$\begin{aligned} 2a(i, j) + 1 &= 4ij + 2i + 2j + 1 \\ &= (2i + 1)(2j + 1). \end{aligned}$$

This explains why the numbers in $\mathcal{B}$ are just the odd composite numbers.

So what Sundaram is doing is to generate the odd composite numbers in a rather unusual way!

It follows that *all odd numbers greater than 1 that do not belong to $\mathcal{B}$ are prime numbers*. The complete list of odd primes can be generated by this procedure.

Python code for Sundaram's Sieve

Source: <https://www.geeksforgeeks.org/sieve-sundaram-print-primes-smaller-n/>

```
# Python3 program to print
# primes smaller than n using
# Sieve of Sundaram.
```

```
# Prints all prime numbers smaller
def SieveOfSundaram(n):

    # In general Sieve of Sundaram,
    # produces primes smaller
    # than  $(2 \times x + 2)$  for a number
    # given number x. Since we want
    # primes smaller than n, we
    # reduce n to half
    nNew = int((n - 1) / 2);

    # This array is used to separate
    # numbers of the form  $i+j+2ij$ 
    # from others where  $1 \leq i \leq j$ 
    # Initialize all elements as not marked
    marked = [0] * (nNew + 1);

    # Main logic of Sundaram. Mark all
    # numbers of the form  $i + j + 2ij$ 
    # as true where  $1 \leq i \leq j$ 
    for i in range(1, nNew + 1):
        j = i;
        while((i + j + 2 * i * j) <= nNew):
            marked[i + j + 2 * i * j] = 1;
            j += 1;

    # Since 2 is a prime number
    if (n > 2):
```

```
print(2, end = " ");

# Print other primes. Remaining
# primes are of the form 2*i + 1
# such that marked[i] is false.
for i in range(1, nNew + 1):
    if (marked[i] == 0):
        print((2 * i + 1), end = " ");

# Driver Code
n = 40;
SieveOfSundaram(n);

# This code is contributed by mits
```

Sample output

On running the above code we get the following output:

```
2 3 5 7 11 13 17 19 23 29 31 37
=== Code Execution Successful ===
```

Article contributed by Shailesh Shirali

Heron's Formula for Area of a Triangle – 1

Heron's formula for the area of a triangle is well-known as it is part of the regular school curriculum. Students would surely have wondered how the early mathematicians came upon this formula; there is no easy-to-understand justification for it. We describe three approaches here that culminate in Heron's formula. The first one is well-known but we include it here for the sake of completeness. The other two approaches are much less known but hold great interest. One of these is due to Heron himself.

For some information about the geometer Heron of Alexandria, please refer to page 98.

We first describe the 'obvious' approach to proving the formula. All it requires is knowledge of the following:

- Pythagoras theorem (also called the 'hypotenuse theorem' or the '*Baudhayana* theorem');
- The difference of two squares formula: $a^2 - b^2 = (a - b)(a + b)$.

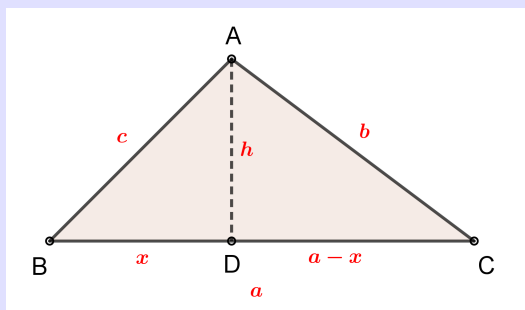


Figure 28

We are given a triangle ABC . We wish to find its area in terms of its sides a, b, c . At least two angles of the triangle must be acute; let's suppose that $\angle B$ and $\angle C$ are acute. Let the configuration be as shown in Figure 1. Let AD be the perpendicular from A to BC (note that D will lie between B and C because angles $\angle B$ and $\angle C$ are both acute), and let the length of AD be h . The area of $\triangle ABC$ is $\Delta = \frac{1}{2}ah$. So, if we can express h in terms of a, b, c , then we will obtain the desired formula for the area. Let $BD = x$, $CD = a - x$.

We have, from the right-angled triangles ADB and ADC :

$$c^2 - x^2 = h^2,$$

$$b^2 - (a - x)^2 = h^2,$$

$$\therefore c^2 - x^2 = b^2 - (a - x)^2$$

This yields on simplification:

$$\begin{aligned} 2ax &= a^2 - b^2 + c^2, \\ \therefore x &= \frac{a^2 - b^2 + c^2}{2a}. \end{aligned} \quad (16)$$

We are now in a position to compute h using the relation $h^2 = c^2 - x^2$. We have:

$$\begin{aligned} h^2 &= c^2 - x^2 \\ &= c^2 - \left(\frac{a^2 - b^2 + c^2}{2a} \right)^2 \\ &= \left(c - \frac{a^2 - b^2 + c^2}{2a} \right) \cdot \left(c + \frac{a^2 - b^2 + c^2}{2a} \right). \end{aligned} \quad (17)$$

Hence:

$$4a^2h^2 = (2ac + a^2 - b^2 + c^2) \cdot (2ac - a^2 + b^2 - c^2). \quad (18)$$

We now examine each bracketed expression separately:

$$\begin{aligned} 2ac + a^2 - b^2 + c^2 &= (a^2 + 2ac + c^2) - b^2 \\ &= (a + c)^2 - b^2 \\ &= (a + b + c)(a + c - b). \end{aligned} \quad (19)$$

Similarly:

$$\begin{aligned} 2ac - a^2 + b^2 - c^2 &= b^2 - (a^2 - 2ac + c^2) \\ &= b^2 - (a - c)^2 \\ &= (b - a + c)(b + a - c). \end{aligned} \quad (20)$$

Therefore we have:

$$4a^2h^2 = (a+b+c)(a+c-b)(b-a+c)(b+a-c). \quad (21)$$

Since $\Delta = \frac{1}{2}ah$, we get $4a^2h^2 = 16\Delta^2$.

Bringing together all the strands, we obtain the desired formula for the area:

$$16\Delta^2 = (a+b+c)(a+c-b)(b-a+c)(b+a-c). \quad (22)$$

This elegant result is known as Heron's formula. It may be put in a visually nicer form by using the symbol for semi-perimeter of the triangle, s :

$$s = \frac{a+b+c}{2}. \quad (23)$$

This yields:

$$a+b+c = 2s, \quad a+c-b = 2s-2b,$$

$$b-a+c = 2s-2a, \quad b+a-c = 2s-2c,$$

hence $16\Delta^2 = 16s(s-a)(s-b)(s-c)$. This yields a truly beautiful result:

$$\Delta^2 = s(s-a)(s-b)(s-c). \quad (24)$$

It is important that the algebra does not hide the simplicity of this proof. Though the manipulations seem lengthy, they are of a simple nature; all we have used is Pythagoras' theorem and the 'difference-of-two-squares' formula — two well-known and powerful results. ■

Symmetry in Heron's formula

The reader may be familiar with the notion of symmetry in geometry but not in algebra.

Consider the expressions $a + b$ and ab . If we swap a and b , the expressions get transformed to $b + a$ and ba . Since addition and multiplication are commutative, the new expressions are essentially the same as the original ones. We can schematically depict this symmetry as follows:

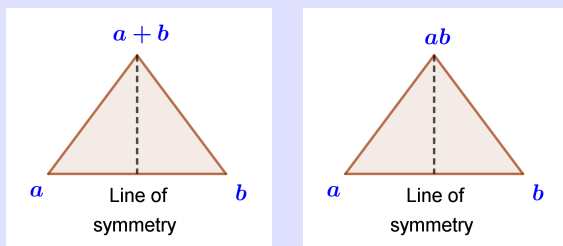


Figure 29: Depicting the symmetry in the expressions $a + b$ and ab

Now consider the expressions $a - b$ and $a + b^2$. If we swap a and b the expressions get transformed to $b - a$ and $b + a^2$. Now in general we have $b - a \neq a - b$ and $b + a^2 \neq a + b^2$. (Equality may hold for some particular values of a and b . In general, however, equality does not hold.)

We say that $a + b$ and ba are **symmetric in a and b** , while $a - b$ and $a + b^2$ are **not symmetric in a and b** .

Similarly for expressions in three or more variables. Thus, the expression $ab + bc + ca$ is symmetric in a, b, c , as its value is unchanged if you shuffle a, b, c among themselves; but the expression $a + b^2 + c^3$ is not symmetric in a, b, c .

In passing we make the following observation about symmetric expressions. Given two symmetric expressions in some given variables, we may combine them in any way we wish and the result will once again be a symmetric expression in those variables.

It stands to reason, surely, that the formula for the area of a triangle with sides a, b, c must be symmetric in a, b, c . It would be odd indeed if the area depended on b differently from the way it depended on a . Examining (22), we see that Heron's formula is indeed symmetric in a, b, c .

Heron's formula can be written in other, equally attractive ways that bring out its symmetry, e.g.:

$$16\Delta^2 = 2(a^2b^2 + b^2c^2 + c^2a^2) - (a^4 + b^4 + c^4), \quad (25)$$

and:

$$16\Delta^2 = (a^2 + b^2 + c^2)^2 - 2(a^4 + b^4 + c^4). \quad (26)$$

Heron's formula and the triangle inequality

Say we are given the following problem in geometry: **Find the area of the triangle whose sides are 1, 2 and 5**. Substituting $a = 1, b = 2, c = 5$ in (22), we get:

$$16\Delta^2 = 8 \times 3 \times 2 \times (-1),$$

leading to $\Delta^2 = -48/16 = -3$.

But Δ^2 is the square of a real number and therefore cannot be negative.

So a triangle with sides 1, 2 and 5 cannot exist at all!

Of course, the reason behind this situation is obvious: *the given lengths do not satisfy the triangle inequality* (we have $1 + 2 < 5$). This means that it is not possible to construct a triangle with sides 1, 2, 5.

This phenomenon will occur whenever we have three positive numbers a, b, c that do not satisfy the triangle inequality. In all such situations, *Heron's formula will yield a negative value for Δ^2* .

Another such situation arises when we consider the following problem: **Find the area of the triangle whose sides are 1, 2 and 3.** Substituting $a = 1$, $b = 2$, $c = 3$ in (22), we get:

$$16\Delta^2 = 6 \times 2 \times 1 \times 0,$$

leading to $\Delta^2 = 0$. So the area of the triangle is 0. This result has come about because the sum of two sides is equal to the third side ($1 + 2 = 3$).

A triangle with zero area is known as a **degenerate triangle**. If we try to construct such a triangle, we find that the triangle collapses into a line segment and therefore has zero area. So the result is not unexpected at all; it makes complete sense.

It follows from the above discussion that *Heron's formula "contains" within it the triangle inequality*. Note what we have found: a connection between two important results in geometry.

Remarks on dimensionality

Let's look at Heron's formula from the point of view of dimensionality:

$$16\Delta^2 = (a + b + c)(a + c - b)(b - a + c)(b + a - c).$$

Note the expression on the RHS. It is a product of four terms in each of which all individual terms have degree 1. Hence, the expression, when expanded out, will be a sum of several terms each with degree 4. On the LHS, we have an expression involving the area (Δ). Since area is a quantity with degree 2, it follows that area^2 will be a quantity with degree 4. So the formula is dimensionally correct.

This kind of perspective which is based on dimensionality is often used in physics but not so often in mathematics. But such considerations help in eliminating possibilities that do not make sense.

So, if the following expressions are claimed to yield the area of a triangle with sides a, b, c (we can see the newspaper headline: “New discovery in geometry!”), we can rule them out from considerations of dimensionality or symmetry (or both):

- $\text{Area} = a^3 + b^3 + c^3$ (the expression is dimensionally incorrect though it is symmetric in a, b, c)
- $\text{Area} = a^2 + bc$ (the expression is dimensionally correct but it is not symmetric in a, b, c)
- $\text{Area} = a^2 + \sqrt{b^4 + c^4}$ (the expression is dimensionally correct but it is not symmetric in a, b, c)

Of course, we can easily produce expressions that are dimensionally correct and also symmetric in a, b, c but which do not yield the area of the triangle; e.g., $\sqrt{a^4 + b^4 + c^4}$. So, such arguments can be used to eliminate possibilities but not to prove that some possibility is correct. That is, they can be used to prove wrongness but not rightness!

(If we conclude on the basis of symmetry and dimensionality that the formula $\sqrt{a^4 + b^4 + c^4}$ correctly yields the area of a triangle with sides

a, b, c , the error we are making would be classified as ‘false positive.’) The same is true in physics.

In the next part of this article, we shall present other (very different) proofs for Heron’s formula.

Further reading

- [1] Christy Williams, Crystal Holcomb, and Kayla Gifford, "Heron's Formula for Triangular Area." <https://www.ms.uky.edu/~corso/teaching/math330/heron.pdf>
- [2] Wikipedia, "Heron's formula." https://en.wikipedia.org/wiki/Heron's_formula

Article contributed by Shailesh Shirali

Closing note: Heron of Alexandria

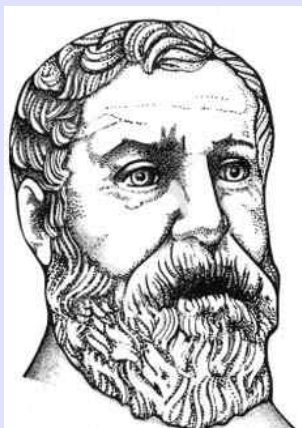


Figure 30: Heron, the great geometer and engineer from Alexandria. Source of image: <https://tinyurl.com/yvzf3lu33>

The following extract has been taken verbatim from <https://tinyurl.com/2nt4ujtw>.

“Heron or Hero of Alexandria was an important geometer and worker in mechanics who invented many machines including a steam turbine. His best known mathematical work is the formula for the area of a triangle in terms of the lengths of its sides. ... From Heron’s writings it is reasonable to deduce that he taught at the Museum in Alexandria. His works look like lecture notes from courses he must have given there on mathematics, physics, pneumatics, and mechanics. Some are clearly textbooks while others are perhaps drafts of lecture notes not yet worked into final form for a student textbook. ... A large number of works by Heron have survived, although the authorship of some is disputed. ... The works fall into several categories, technical works, mechanical works and mathematical works.”

Counting integer-sided triangles

Recently we came across the following problem (it had been posed in one of the mathematical Olympiads in Singapore):

Problem. *How many distinct (non-congruent) triangles are there with sides of integer length, such that the longest side has length 10 units?*

We must clarify that “longest side has length 10 units” includes the possibility that there may be two or more sides with length 10 units.

It is relatively easy to arrive at the answer. But it is much more interesting to consider the problem in greater generality. We shall therefore consider the following:

Problem. *How many distinct (non-congruent) triangles are there with sides of integer length, such that the longest side has length $2n$ units?*

We shall show that the problem has a neat answer: $n(n+1)$.

For example, let $n = 2$. We need to list all possible triangles with sides of integer length, such that the longest side has length 4. The list is clearly the following:

$$(1, 4, 4), (2, 3, 4), (2, 4, 4), (3, 3, 4), (3, 4, 4), (4, 4, 4).$$

We see that there are 6 such triangles.

Or take $n = 3$. The list is now:

$$(1, 6, 6), (2, 5, 6), (2, 6, 6), (3, 4, 6), (3, 5, 6), (3, 6, 6)$$

$$(4, 4, 6), (4, 5, 6), (4, 6, 6), (5, 5, 6), (5, 6, 6), (6, 6, 6)$$

There are 12 such triangles.

In general there will be:

- 1 triangle with smallest side 1, namely, $(1, 2n, 2n)$;
- 2 triangles with smallest side 2, namely, $(2, 2n - 1, 2n)$ and $(2, 2n, 2n)$;
- 3 triangles with smallest side 3, namely, $(3, 2n - 2, 2n)$, $(3, 2n - 1, 2n)$ and $(3, 2n, 2n)$;
-
- n triangles with smallest side n , namely, $(n, n + 1, 2n)$, $(n, n + 2, 2n)$, ..., $(n, 2n, 2n)$;
- n triangles with smallest side $n + 1$, namely, $(n + 1, n + 1, 2n)$, $(n + 1, n + 2, 2n)$, ..., $(n + 1, 2n, 2n)$;
- $n - 1$ triangles with smallest side $n + 2$, namely, $(n + 2, n + 2, 2n)$, $(n + 2, n + 3, 2n)$, ..., $(n + 2, 2n, 2n)$; ...

- 2 triangles with smallest side $2n - 1$, namely, $(2n - 1, 2n - 1, 2n)$ and $(2n - 1, 2n, 2n)$;
- 1 triangle with smallest side $2n$, namely, $(2n, 2n, 2n)$.

So the required number of triangles is

$$2(1 + 2 + 3 + \cdots + n) = n(n + 1).$$



Variation of the above problem

At the start we had specified that we would allow more than one side to have the maximum permitted length.

What happens if we do not permit this? The question we ask now is:

Problem. *How many distinct (non-congruent) triangles are there with sides of integer length, such that the longest side has length 10 units, and only one side can have this length?*

And more generally:

Problem. *How many distinct (non-congruent) triangles are there with sides of integer length, such that the longest side has length $2n$ units, and only one side can have this length?*

We will leave it to the reader to explore this problem and to solve it fully. You can be sure that the final answer will be just as neat as the above!

Article contributed by Shailesh Shirali

Problems for Solution – 1

Here are some problems for you to ponder over. The answers are given on page 106. Solutions will be given in the next issue of the magazine. Enjoy!

1. Are there any non-zero perfect squares that have two or more digits and are of the form $aaa\dots a$? (Numbers are represented in base 10.)
2. Find the smallest perfect square whose digits start with 1234567. (That is, the number is of the form $1234567\dots$ where the dots represent more digits. Numbers are represented in base 10.)
3. How many distinct integers n are there (positive or negative) such that $n^2 + 20n + 24$ is a perfect square?
4. Which is larger,

$$1.0001 + \frac{1}{1.0001} \quad \text{or} \quad 1.001 + \frac{1}{1.001}?$$

Try to answer this without reaching for the calculator!

5. Which is larger, $200!$ or 100^{200} ?

Answers to “Crazy Cryptarithms” – 1

We give here the answers to the cryptarithms on page 18. Note that these are only the answers; we will give the solutions in the next issue.

1. To solve: $ABCD \times 4 = DCBA$.

Answer: $2178 \times 4 = 8712$.

2. To solve: $AB + BC + CA = ABC$.

Answer: $19 + 98 + 81 = 198$

3. To solve: $BAD \times 9 = GOOD$, with $A = 1$.

Answer: $815 \times 9 = 7335$

4. To solve: $ROUND \times 3 = SQUARE$.

Answer: $35746 \times 3 = 107238$

Answers to “Problems for Solution” – 1

We give here the answers to the problems on page 104.

Note that these are only the answers; we will give the solutions in the next issue.

1. None. Sadly, there are no perfect squares of the stated type!
2. The least such perfect square is the number 1234567654321. (We will describe in the next issue how one can go about answering such a question.)
3. There are only two such values of n , namely, $n = -30$ and $n = 10$. Both yield the same value (324) for $n^2 + 20n + 24$.
4. $1.001 + 1/1.001$ is larger.
5. 100^{200} is larger.

Call for Articles

We invite mathematics teachers and practitioners to contribute short articles to The Compass.

Articles could deal with mathematical content or the teaching of mathematics (classes 6 to 12). They could feature real-life classroom experiences with projects and explorations.

They could be about the use of audiovisual material or social media in the mathematics classroom.

They could feature reflections on the teaching of mathematics, or on the nature of mathematics.

Articles about the use of software are particularly welcome, as are articles dealing with non-routine problems and classroom experiences with problem-solving.

In short, anything at all to do with school mathematics!